

KISHINCHAND CHELLARAM COLLEGE, MUMBAI – 20.

FY / SY / TY B. Sc (I. T.) Semester_____

NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 1

TEACHER'S SIGNATURE:

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Title: Windows 10 Installation on VMware Workstation Pro

Aim: To install Windows 10 operating system as a guest OS on VMware Workstation Pro to create a virtual machine environment for cloud computing practicals.

Objective: To understand the concept of virtualization and learn how to create and configure a virtual machine, and install a guest operating system on it using VMware Workstation Pro.

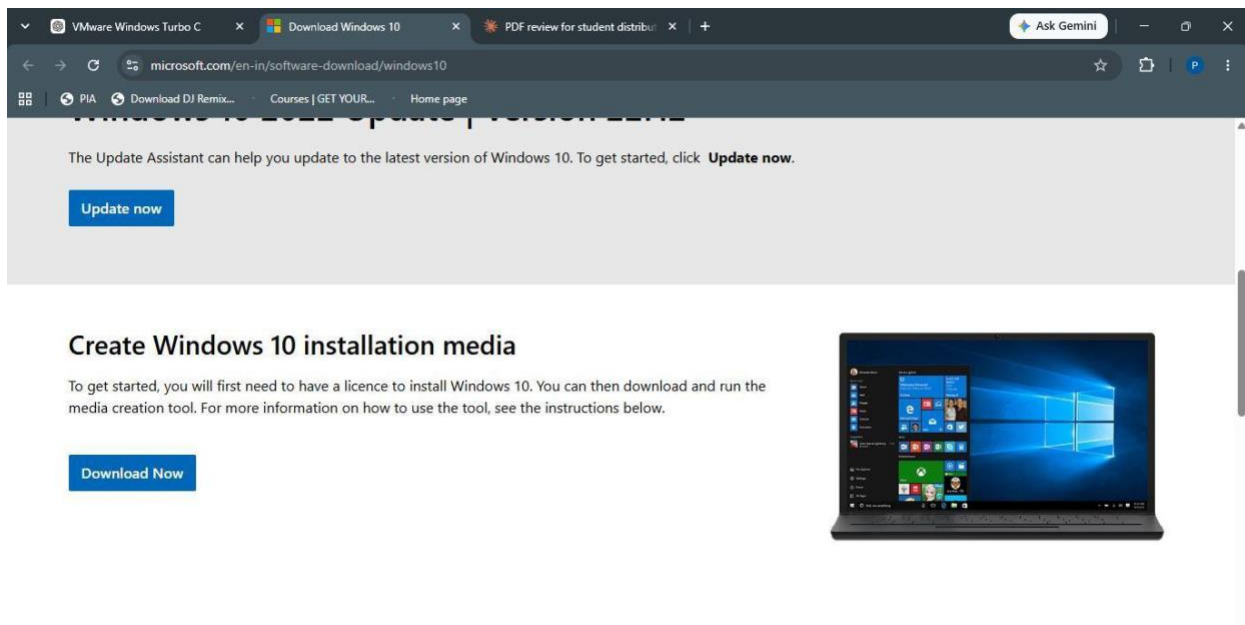
Prerequisites: (Keep good amount of space vacant)

i) At least 60 GB of free disk space on the host, plus roughly 5–6 GB for the downloaded Windows ISO file.

ii) A stable internet connection to download the Windows 10 ISO (large file, may take a while).

Step 1

Open a browser and go to <https://www.microsoft.com/en-in/software-download/windows10>. Click Download Now under 'Create Windows 10 installation media' to download the Media Creation Tool.

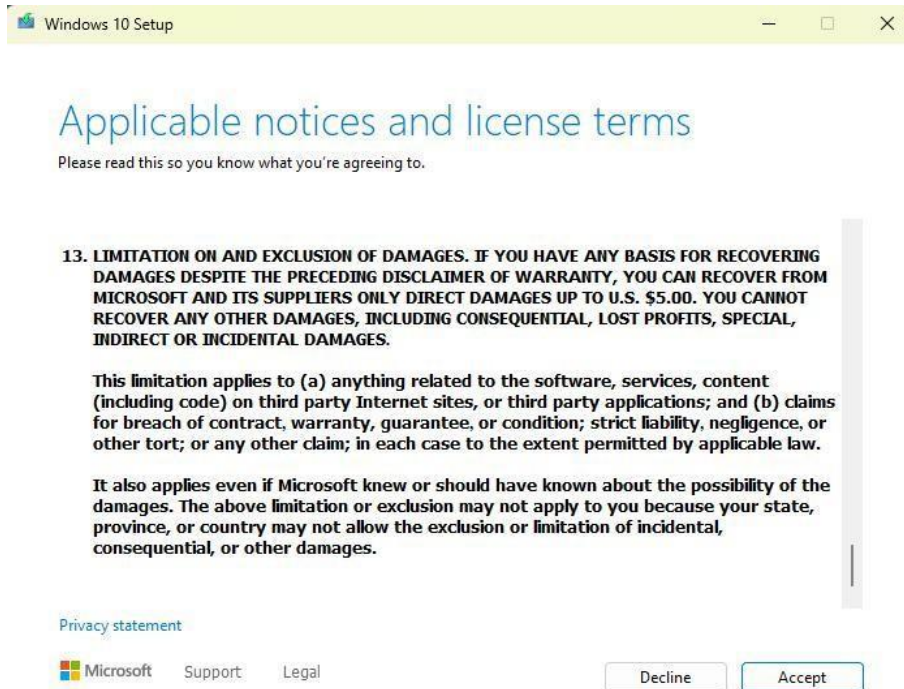


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Step 2

Run the downloaded Media Creation Tool. Read the license terms and click 'Accept' to continue.

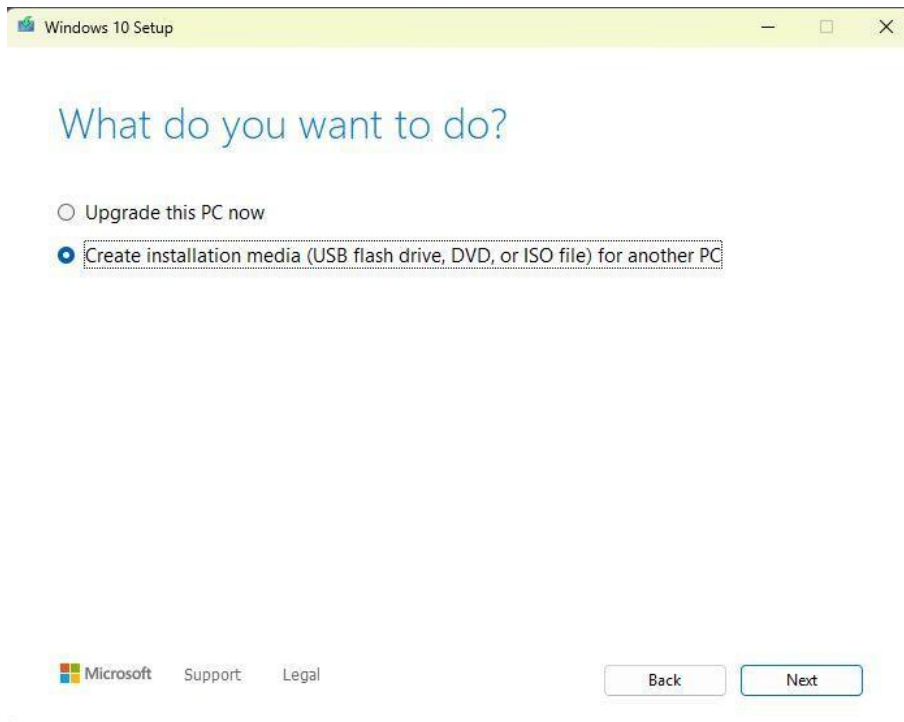


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Step 3

On the 'What do you want to do?' screen, select 'Create installation media (USB flash drive, DVD, or ISO file) for another PC' and click Next. Follow the prompts to choose the language, edition, and architecture (64-bit recommended), then select 'ISO file' as the media to use.

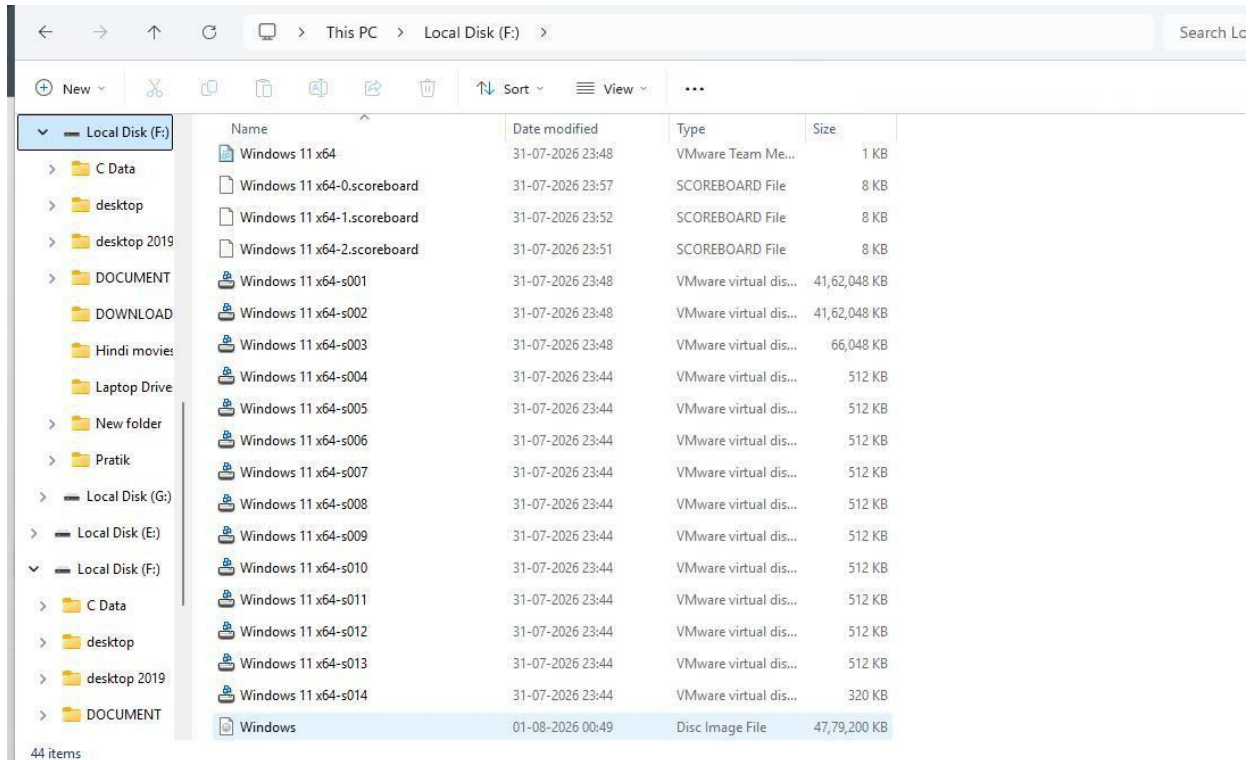


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Step 4

Once the tool finishes, the Windows ISO file will be saved to the folder you chose. Note this location carefully, you will need to browse to it while creating the virtual machine.



Step 5

Open VMware Workstation Pro and click File > New Virtual Machine. Choose 'Typical (recommended)' and click Next. Select 'Installer disc image file (iso)', click Browse, and select the Windows ISO file from Step 4. Click Next.

Step 6

Select Microsoft Windows as the guest operating system and choose Windows 10 x64 (or the version you downloaded) from the version dropdown. Click Next.

Step 7

Enter a name for the virtual machine (e.g. 'Windows 10 x64') and choose the folder where its files will be stored. Click Next.

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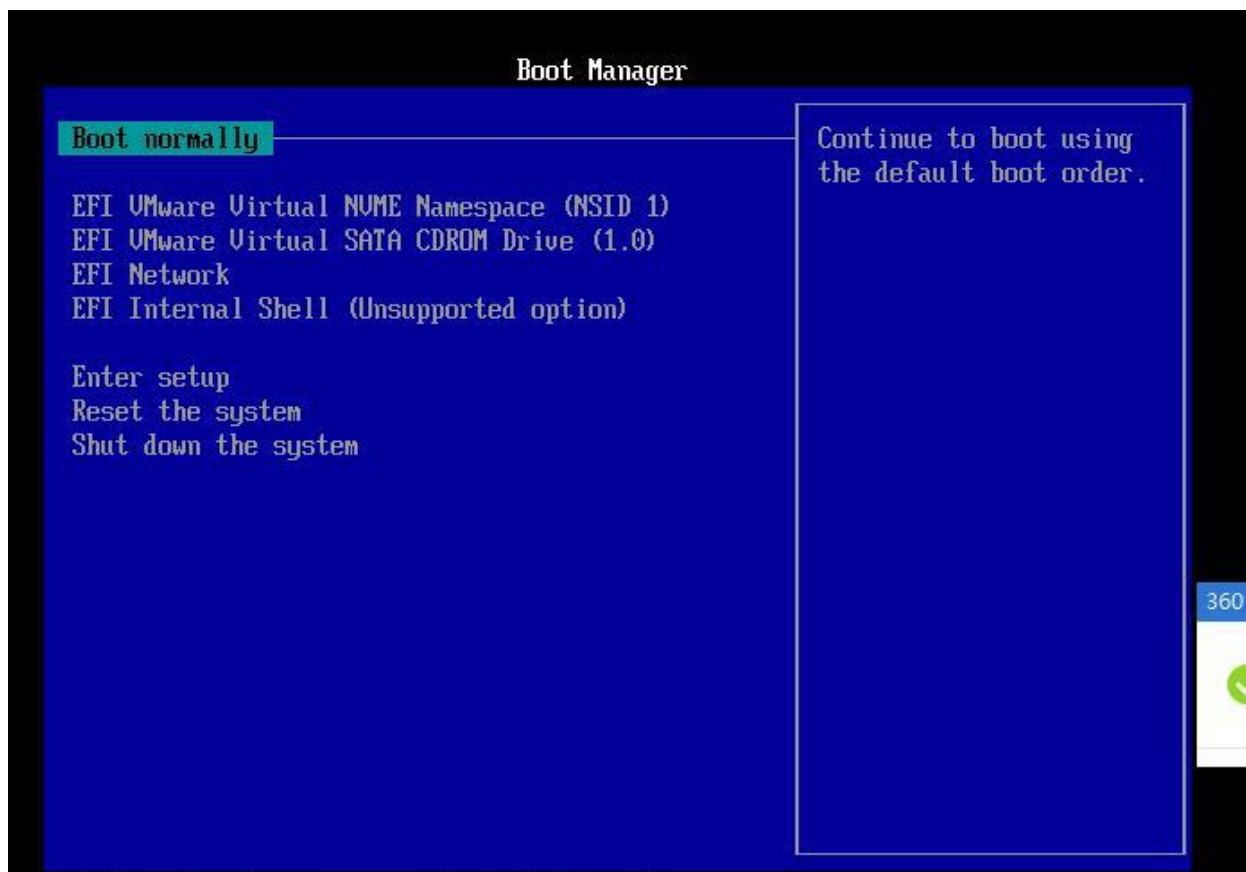
Step 8

Set the maximum disk size (60 GB default is fine) and select 'Store virtual disk as a single file'. Click Next.

Step 9

Review the summary and click Finish. Select the new virtual machine in the Library and click the green Power on this virtual machine button to boot from the ISO.

IF THIS INTERFACE APPEARS, PLEASE SELECT : EFI VMware Virtual SATA CDROM Drive (1.0)

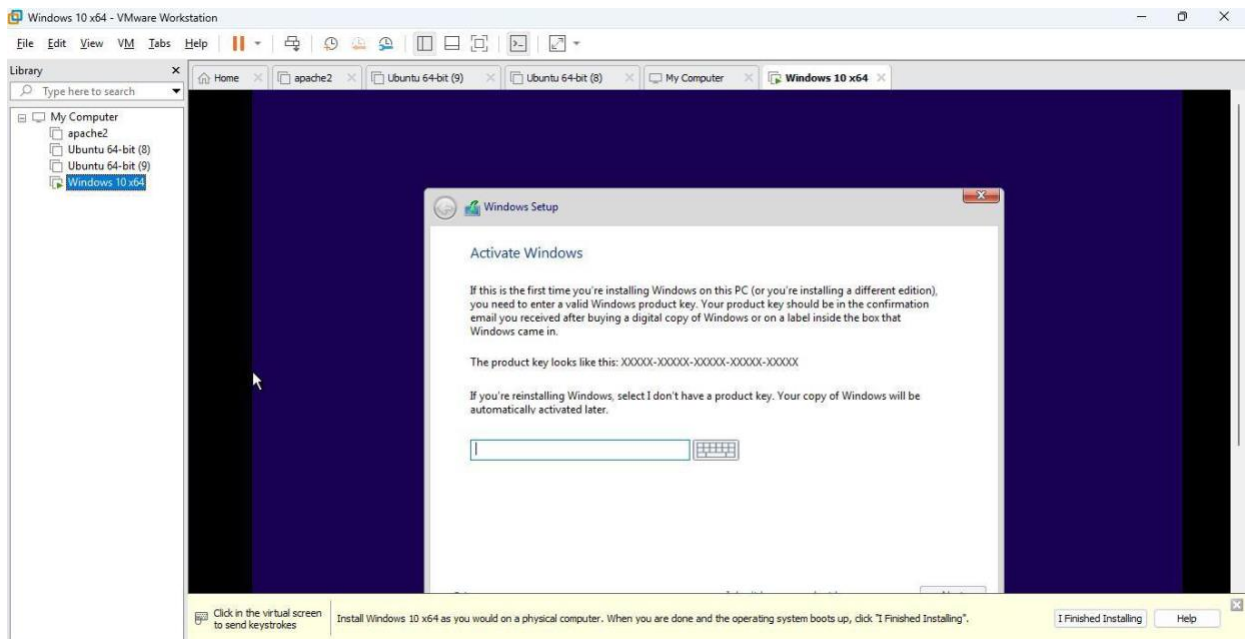


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Step 10

When the 'Activate Windows' screen appears inside the virtual machine, click 'I don't have a product key' at the bottom to continue without activation for now, you can activate later.

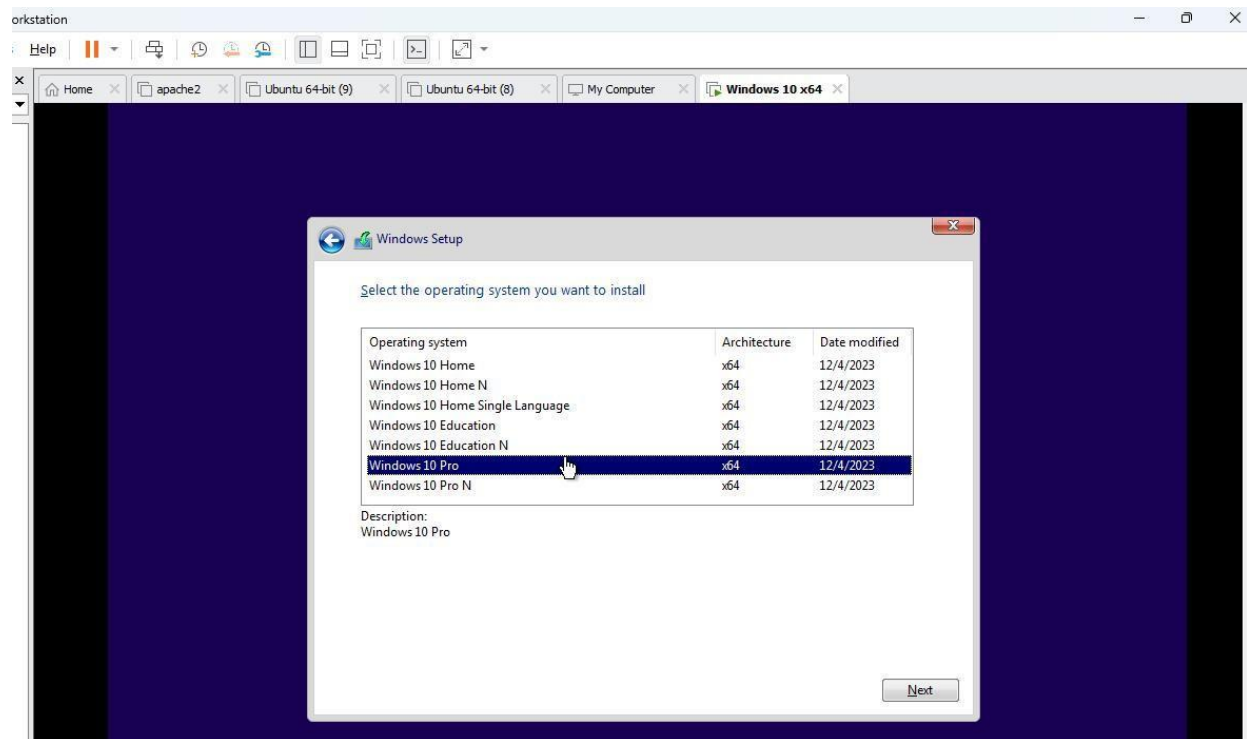


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Step 11

On the Select the operating system you want to install screen, choose the edition your institution has licensed (e.g. Windows 10 Pro) and click Next.



Step 12

Accept the Microsoft license terms, then choose 'Custom: Install Windows only (advanced)'. Select the unallocated virtual disk shown and click Next to begin installation.

Step 13

Windows will copy files, install features, install updates, and restart the virtual machine automatically several times. This can take 15–30 minutes, do not power off the VM during this process.

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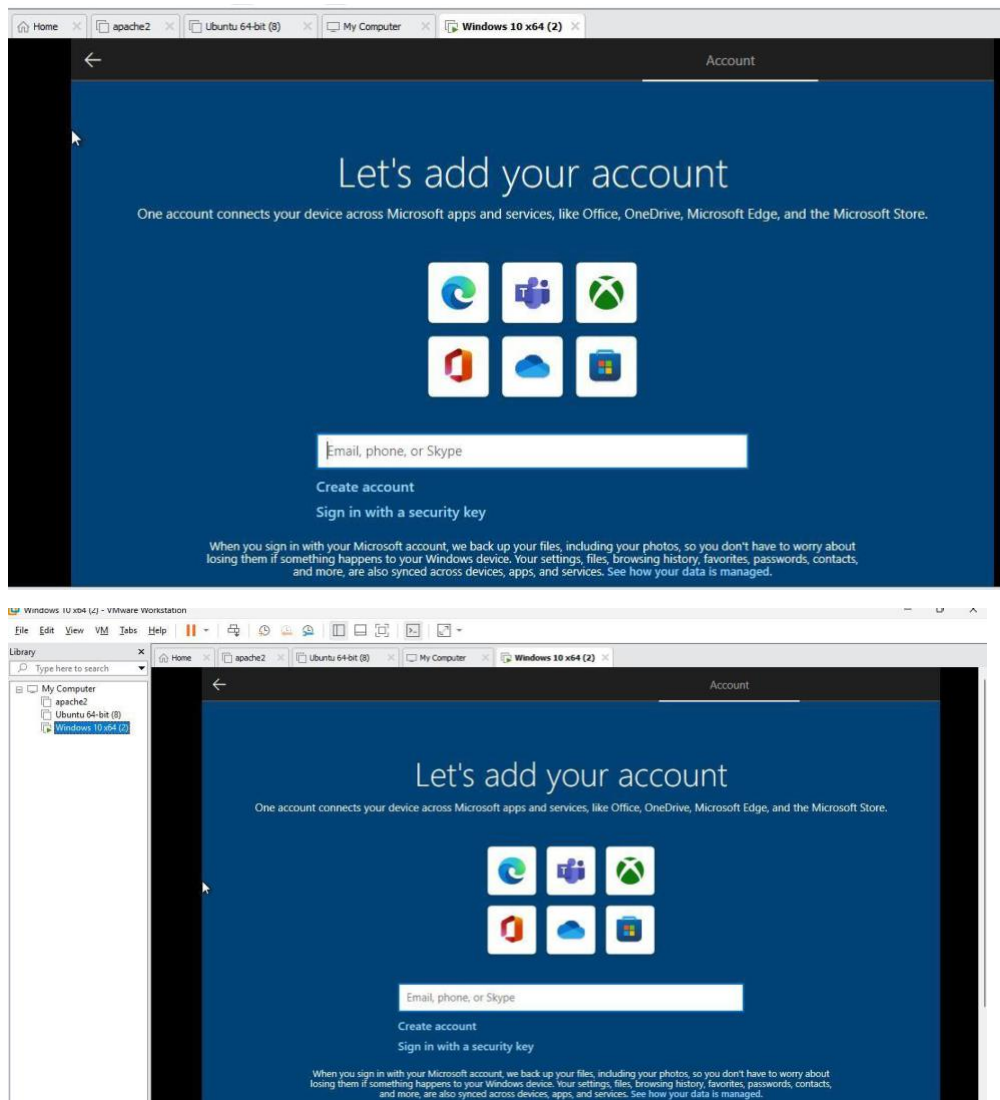
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Step 14

When prompted, confirm your region and keyboard layout by clicking 'Yes' (add a second layout only if you need one).

Step 15

On the 'Let's add your account' screen, sign in with a Microsoft account, or look for a 'Sign-in options' / offline account link to create a local account instead if you don't want to use a Microsoft account.

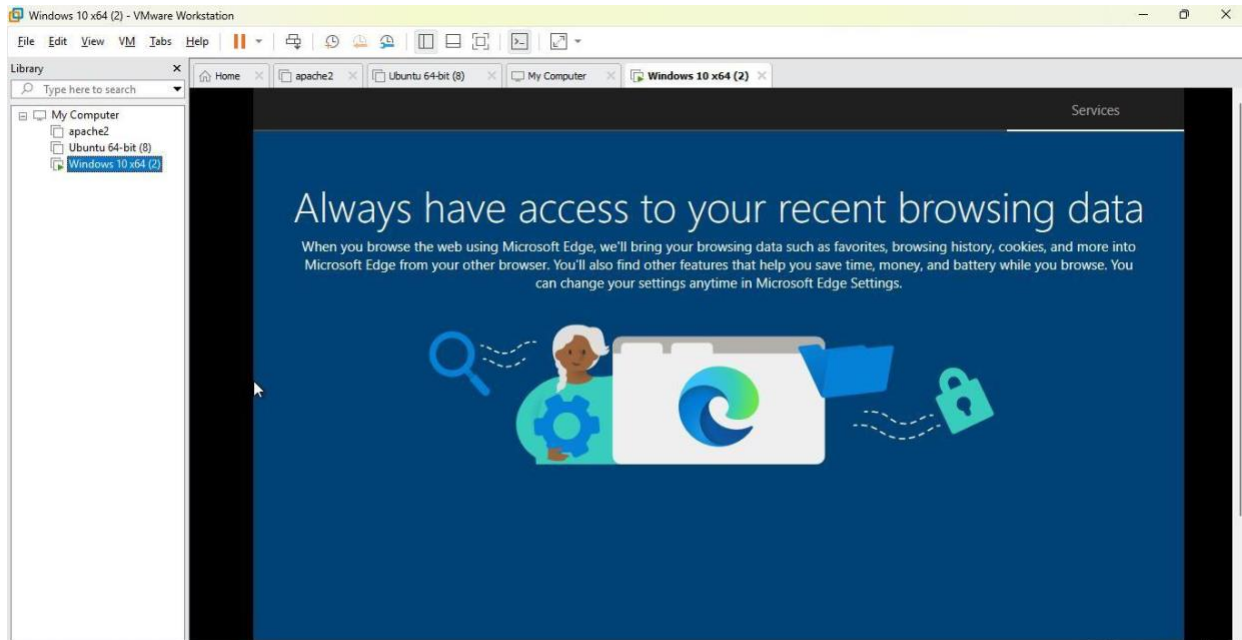


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Step 16

Windows will show a few final setup screens, including one about Microsoft Edge syncing your browsing data. Click 'Accept' or 'Next' to continue.



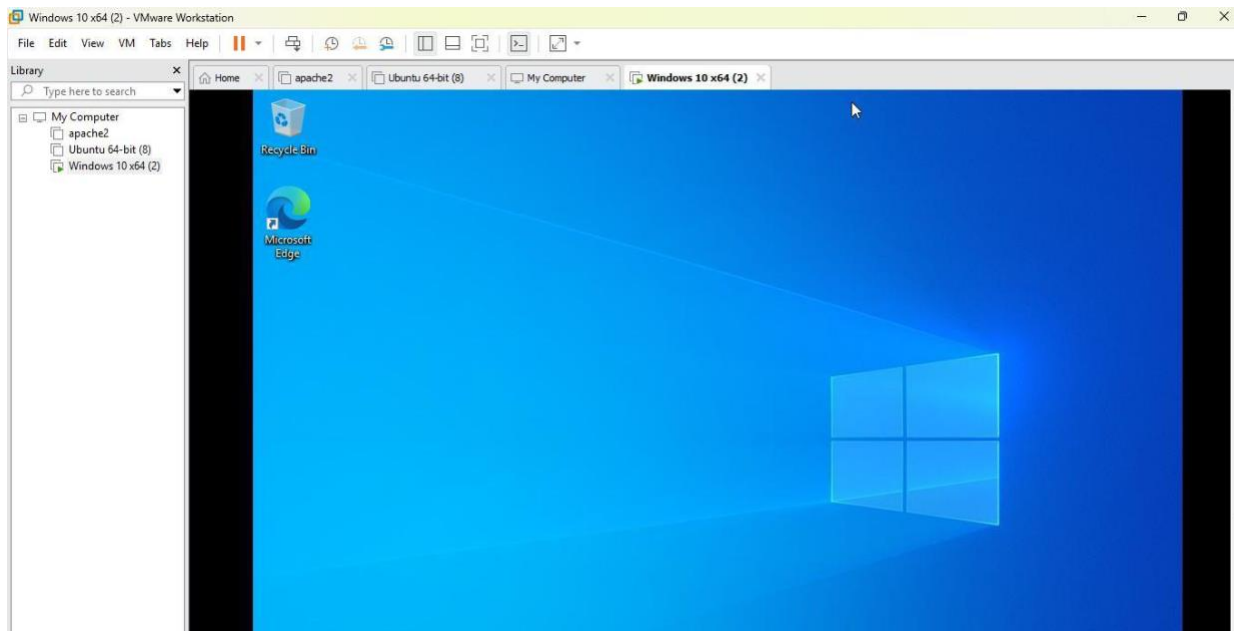
PLEASE MAKE SURE TO ZOOM OUT, TO SEE THE 'ACCEPT' BUTTON

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Result

After setup finishes, the Windows desktop appears with the Recycle Bin and Microsoft Edge icons, Windows 10 has been installed successfully on the virtual machine.



How To Access

You can find your Windows virtual machine anytime in VMware Workstation Pro's Library panel on the left. Select it and click 'Power on this virtual machine' to start it.

Explanation:

- Virtualization allows multiple operating systems to run on a single physical machine using virtual machines (VMs) that act as independent computers.
- VMware Workstation Pro is used as a hypervisor to create a new virtual machine.
- The Windows 10 ISO file is downloaded using the Media Creation Tool and used as the installation media.
- The virtual machine is configured with settings such as disk size, RAM, and processor allocation.
- Windows 10 is installed inside the VM by following the standard installation wizard.
- This VM forms the base environment on which further cloud computing practicals are performed.

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NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 2

TEACHER'S SIGNATURE:

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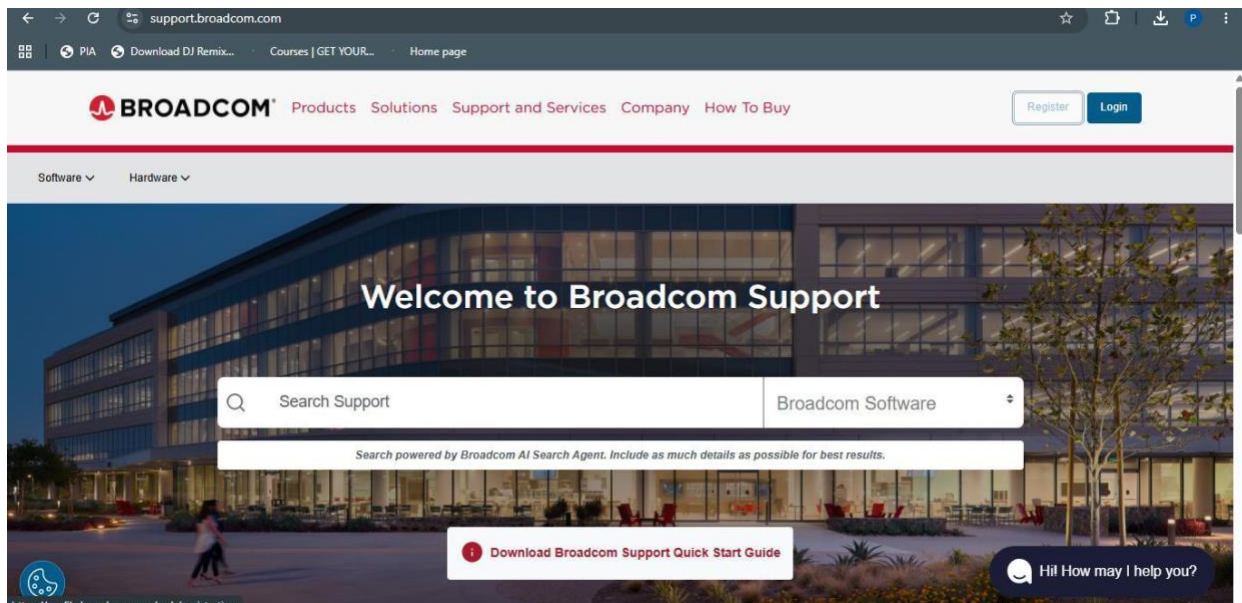
Title: VMware Workstation Pro Installation

Aim: To install VMware Workstation Pro on a Windows machine, which acts as a hypervisor for creating and managing virtual machines.

Objective: To understand the role of a Type-2 hypervisor in cloud computing and to learn the step-by-step procedure of downloading, installing, and verifying VMware Workstation Pro.

Step 1

Open the Broadcom Support website <https://support.broadcom.com/> and register/login.

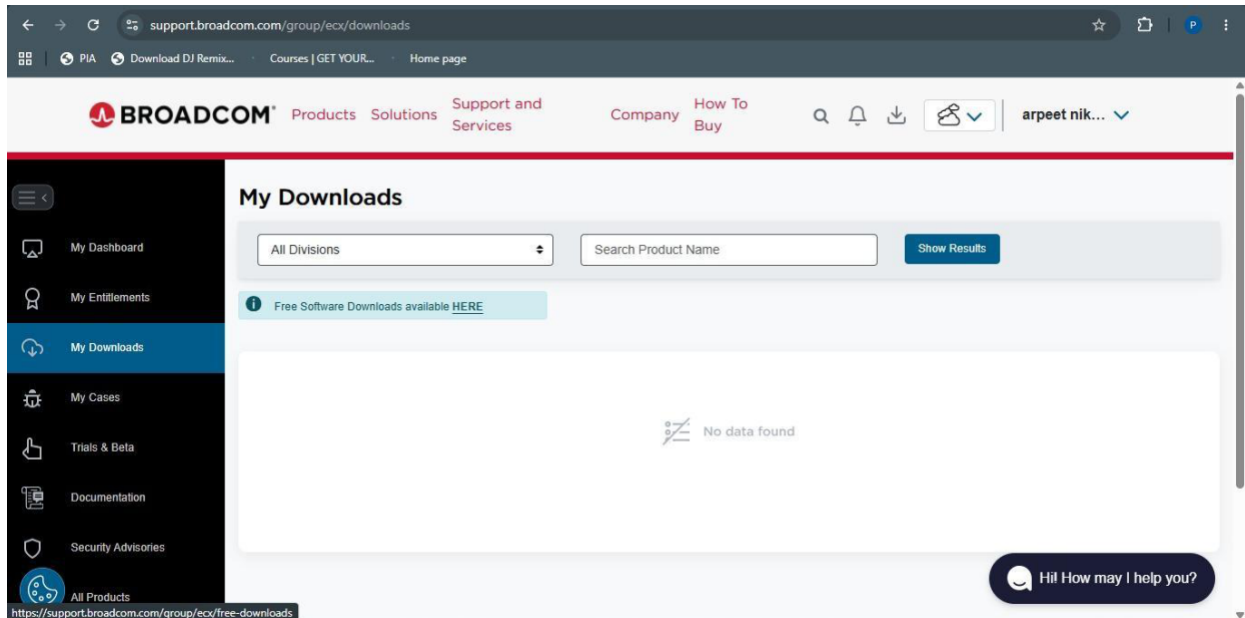


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Step 2

Go to My Downloads. Click the 'HERE' link next to “Free Software Downloads available”.



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Step 3

Search for 'VMware Workstation Pro'.

Free Downloads

All Divisions

vmware workstation pro

Show Results

Reset Search

VMware Workstation Pro

1 to 1 of 1 records

< 1 >

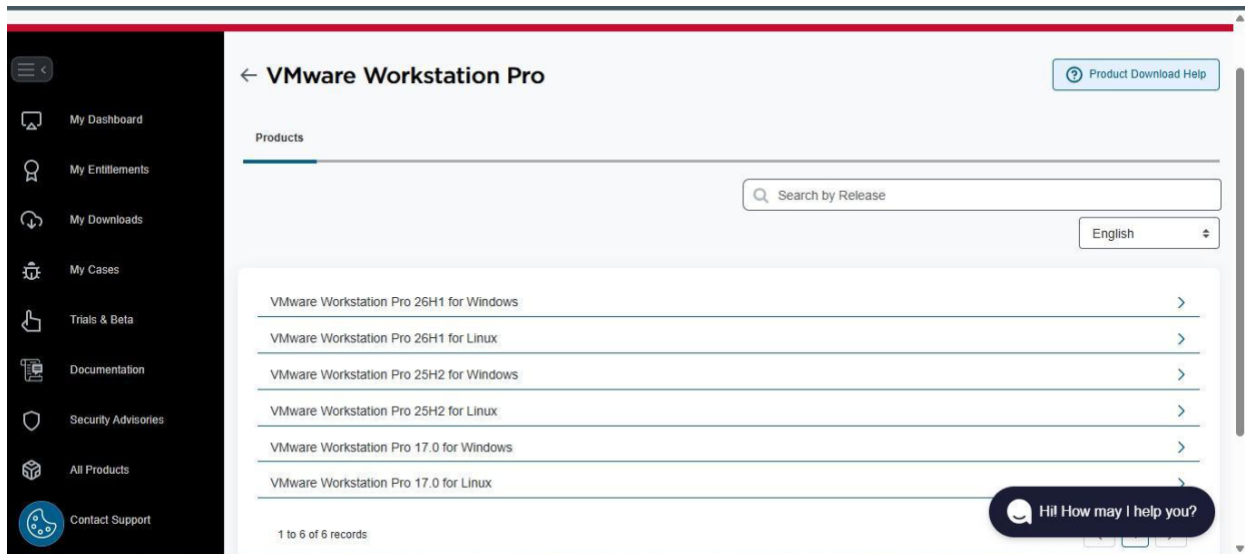
Hit How may I help you?

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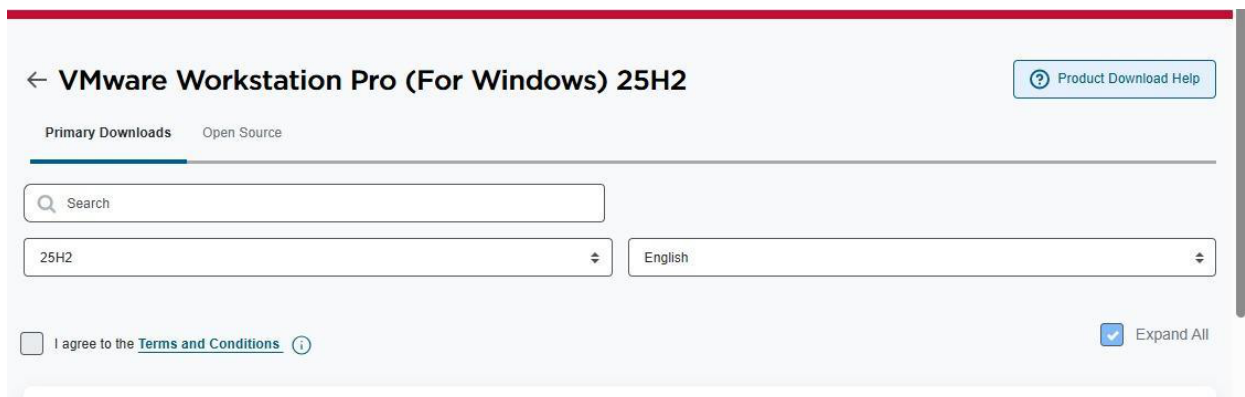
Step 4

Select the required Windows version. Choose the required release (e.g. 26H1/25H2).



Step 5

Click on Terms and Conditions:

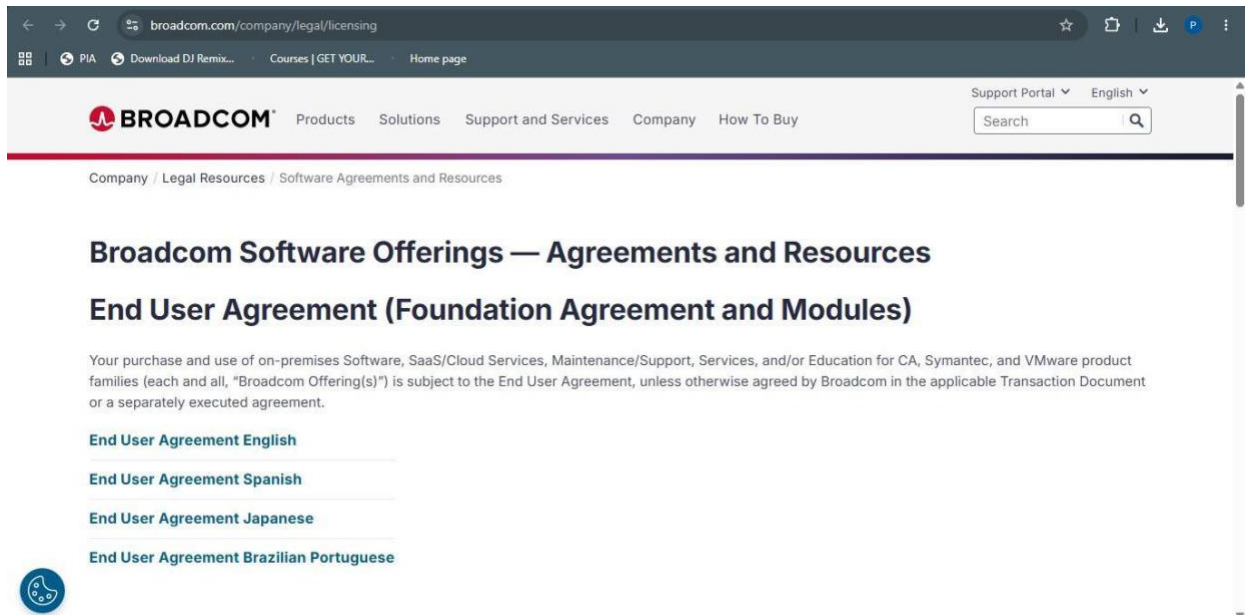


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Step 6

Read the End User Agreement.



The screenshot shows a web browser displaying the Broadcom website. The address bar shows the URL broadcom.com/company/legal/licensing. The page header includes the Broadcom logo and navigation links: Products, Solutions, Support and Services, Company, and How To Buy. A search bar is also present. The main content area is titled "Broadcom Software Offerings — Agreements and Resources" and "End User Agreement (Foundation Agreement and Modules)". Below this, a paragraph states: "Your purchase and use of on-premises Software, SaaS/Cloud Services, Maintenance/Support, Services, and/or Education for CA, Symantec, and VMware product families (each and all, "Broadcom Offering(s)") is subject to the End User Agreement, unless otherwise agreed by Broadcom in the applicable Transaction Document or a separately executed agreement." There are four links for the End User Agreement in different languages: English, Spanish, Japanese, and Brazilian Portuguese. A small circular icon with a question mark is visible in the bottom left corner.

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Step 7

After scrolling through the whole terms document, accept the Terms and Conditions.

Primary Downloads Open Source

Search

25H2 English

☒ I agree to the Terms and Conditions ☒ Expand All

VMware Workstation Pro (For Windows) Release **25H2**

	Release Date	Last Updated	SHA2	MD5	
63	Oct 14, 2025	Oct 09, 2025	49ad7c2bbce854ed30ed0702d1af9fc042697777dc981e087bfa7241045b0361	e9476168dfecaf1359a371c2500de7f8	HTTPS Download

Hi! How may I help you?

Step 8

Click the HTTPS Download icon to download the installer.

My Entitlements My Downloads My Cases Trials & Beta Documentation Security Advisories All Products Contact Support

Search

26H1 543221 English

☒ Expand All

VMware Workstation Pro (For Windows) Release **26H1** Release Level Info **543221**

	Release Date	Last Updated	SHA2	MD5	
34	May 14, 2026	May 05, 2026	a0ef9087607d9cad20b08139e73e41242e044ad5bd8cee141d3bad314586737f	83f9b6cc7cf0e74ad3b15ccca27860af	HTTPS Download

1 to 1 of 1 records

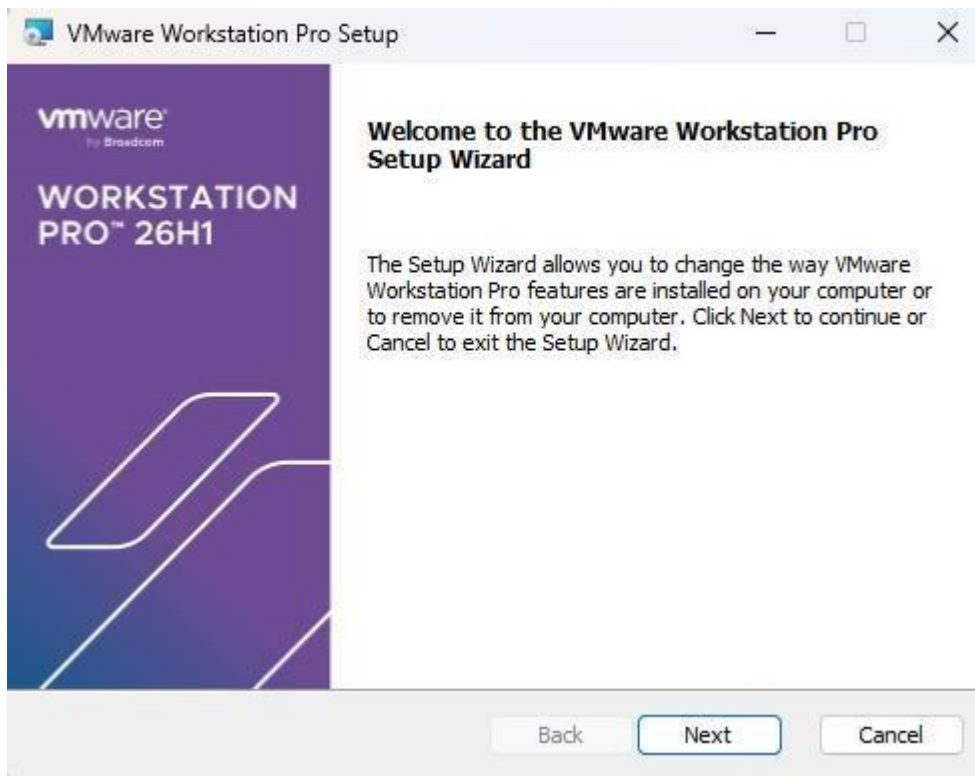
Hi! How may I help you?

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Step 9

Run the installer and click Next. Continue with the default options.

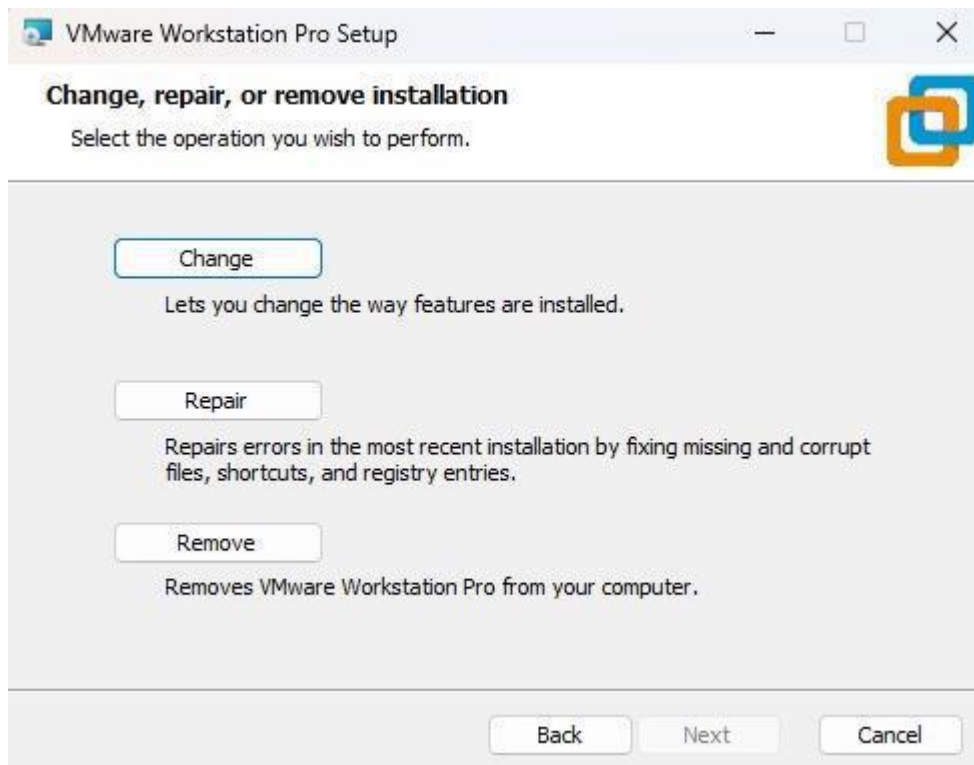


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Step 10 (only if VMware is already installed on this machine)

If VMware is already installed, the Change/Repair/Remove screen will appear. Select Change (if asked). If installing for the first time, you will not see this screen, continue with the default install path shown by the wizard.



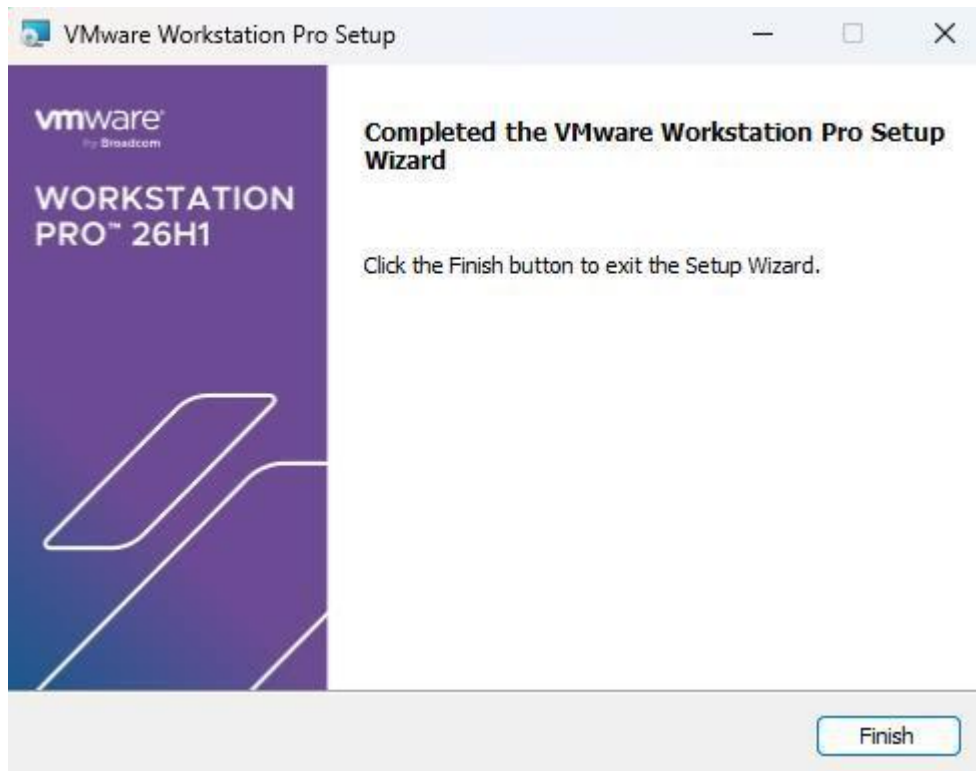
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Result

Click on Finish. VMware Workstation Pro was downloaded and installed successfully on Windows.

You may be prompted to restart your computer to complete the installation , save your work before continuing.



Verify the Installation

Open VMware Workstation Pro from the desktop shortcut or Start menu. You should see the welcome screen with options such as Create a New Virtual Machine. If this screen appears, the installation was successful.

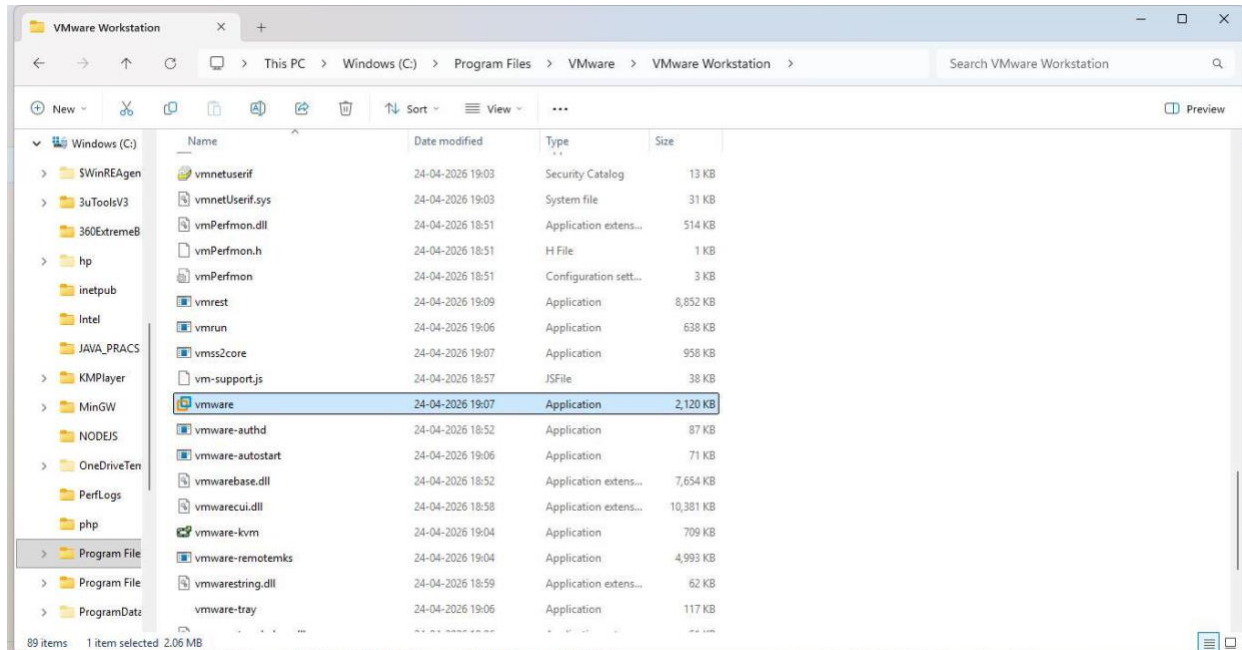
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How To Access

You can access it directly via the Desktop shortcut, or by going to:

C: Drive > Program Files > VMware > VMware Workstation



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If VMware does not open or has any glitch,
TROUBLESHOOT:

Download link not working: Make sure you accepted the Terms and Conditions checkbox (Step 7) before clicking the download icon.

Installer won't launch: Right-click the downloaded .exe file and choose 'Run as administrator'.

Explanation:

- A hypervisor is software that enables virtualization by allowing multiple VMs to run on a single physical host.
- VMware Workstation Pro is a Type-2 (hosted) hypervisor that runs on top of an existing operating system.
- The installer is downloaded from the Broadcom Support website after registering/logging in.
- The End User License Agreement is accepted before proceeding with the download.
- The installer is run and installed on the host machine using the setup wizard with default options.
- Successful installation is verified by launching the application and viewing the welcome screen with the "Create a New Virtual Machine" option.

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NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 3

TEACHER'S SIGNATURE:

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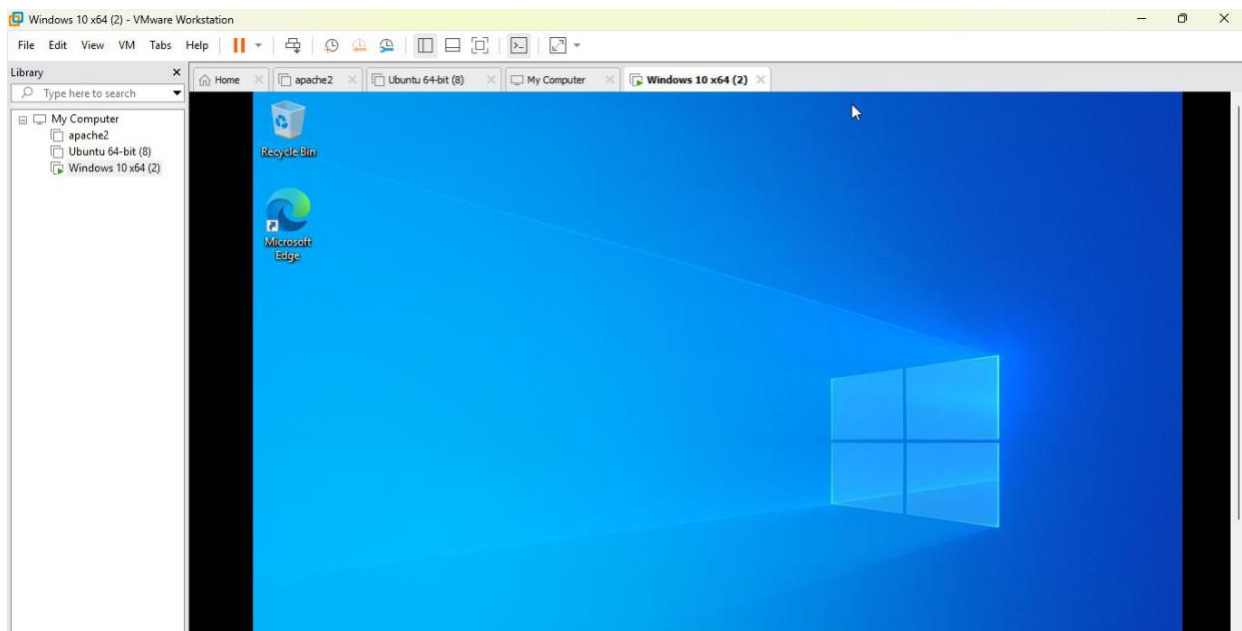
Title: Turbo C++ Installation & Program Execution (Inside the Windows Virtual Machine)

Aim: To install Turbo C++ inside a virtual machine and execute a simple C++ program to verify the working of the virtualized environment.

Objective: To understand how application software can be installed and executed inside a virtual machine, and to verify that the virtualized environment functions like a physical machine.

Step 1

Power on the Windows 10 virtual machine in VMware Workstation Pro and wait for the desktop to load.

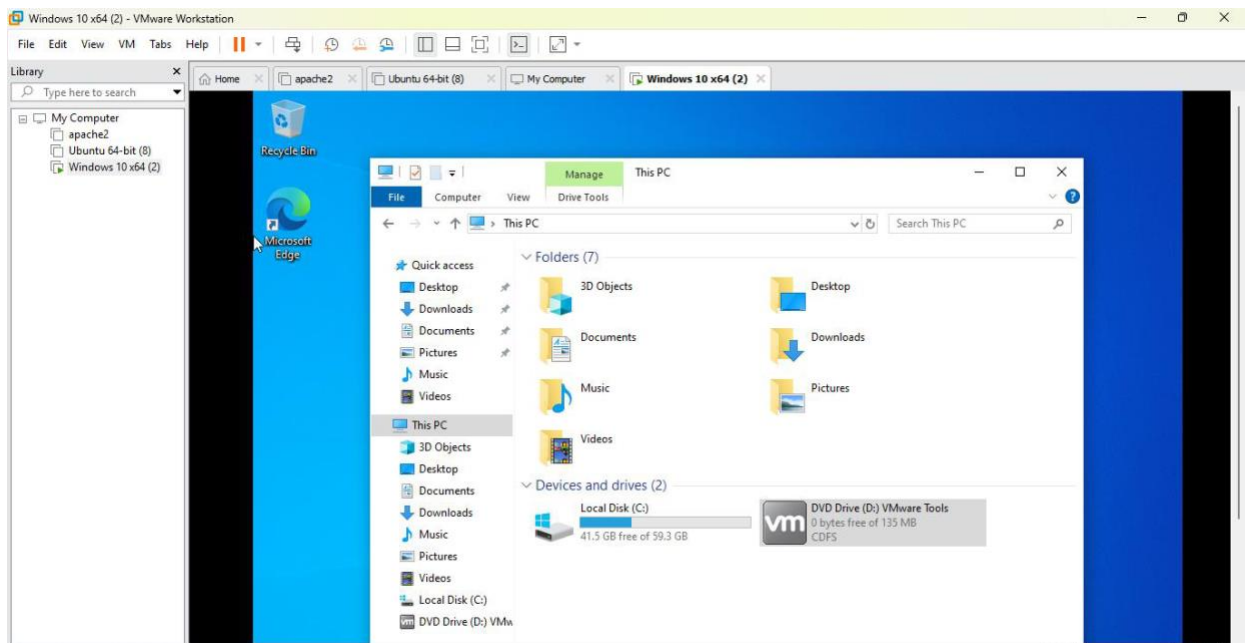


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Step 2

Open File Explorer and go to 'This PC' to check available drives



Click on DVD Drive VMware Tools:

Run setup.exe

Keep clicking Next → Next → Install → Finish

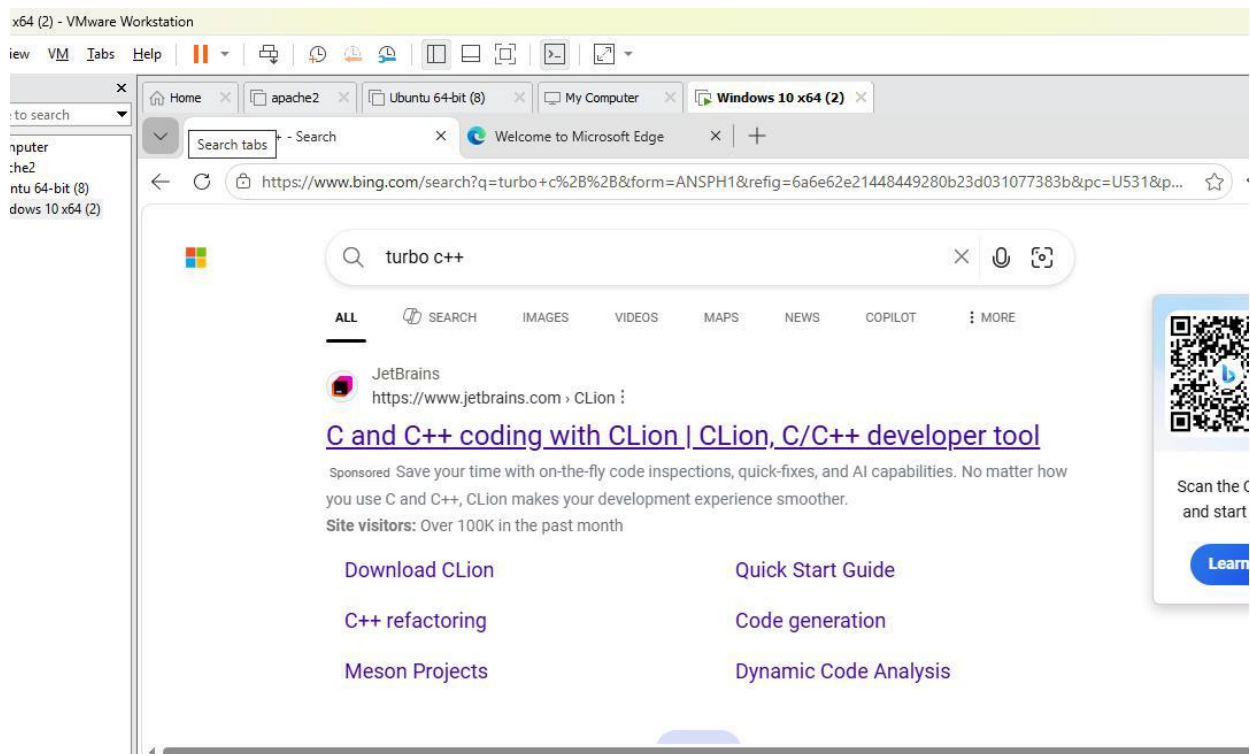
Restart the VM.

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Step 3

Open Microsoft Edge (or any browser) inside the virtual machine and search for Turbo C++ download . You can download from <https://sourceforge.net/> or <https://turbo-c.en.softonic.com/>



Step 4

Once downloaded, right-click the TURBOC3.zip file and choose Extract All. Extract it to a simple path such as C:\TURBOC3 so it's easy to find later.

Step 5

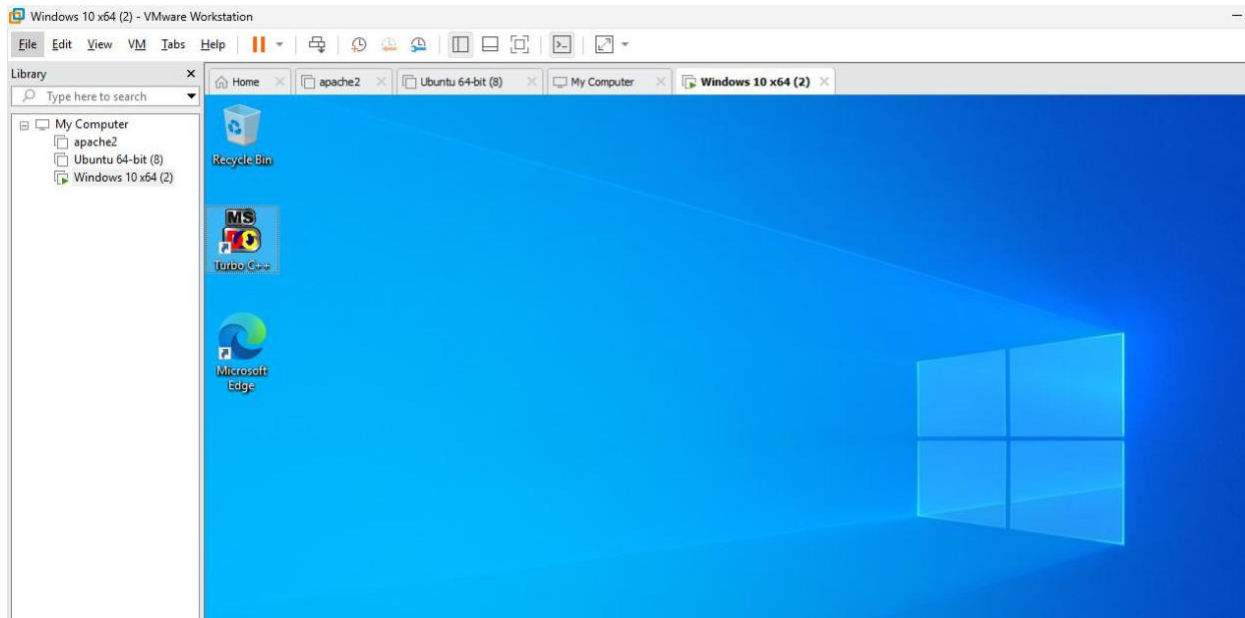
Open the extracted TURBOC3 folder, then open the BIN sub-folder and double-click TC.EXE to launch Turbo C++. Right-click TC.EXE and choose 'Send to > Desktop (create shortcut)' so you can open it quickly next time.

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Step 6

Once set up, a Turbo C++ shortcut icon will be available on the desktop for quick access any time you need it.



Step 7

Inside Turbo C++, go to File > New to open a blank editor, then type the following program:

```
#include<iostream.h>
#include<conio.h>
void main()
{
    clrscr();
    cout<<"hello BSC-IT";
    getch();
}
```

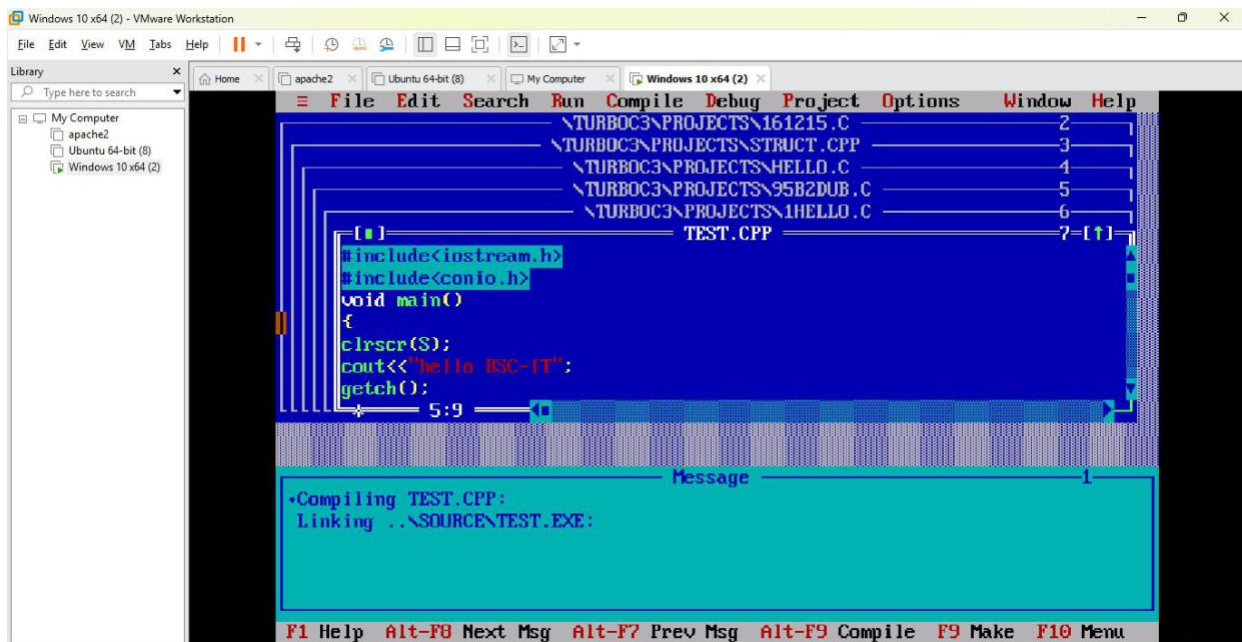
Save the file using F2 (e.g. as TEST.CPP).

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Step 8

Compile the program with Alt + F9.

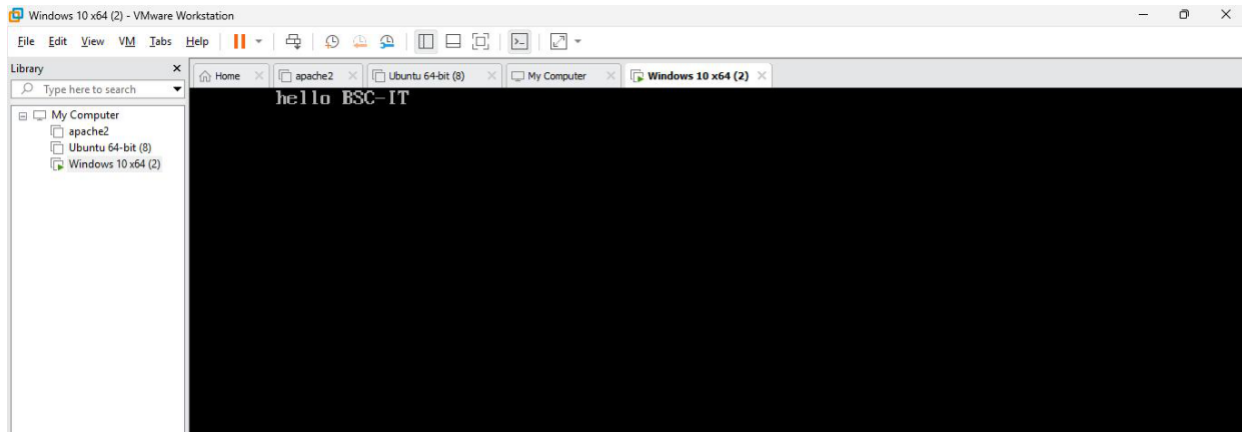


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Step 9

Run the program with Ctrl + F9. The output screen will appear showing the result of the program.



Step 11

Press any key to return to the Turbo C++ editor. To exit Turbo C++ completely, press Alt + X, which takes you back to the Windows desktop.

Result

Turbo C++ was installed successfully inside the Windows virtual machine, and the program compiled and ran correctly, printing 'hello BSC-IT' to the output screen.

How To Access

Open Turbo C++ anytime using the desktop shortcut, or by navigating to C:\TURBOC3\BIN\TC.EXE.

Explanation:

- VMware Tools is installed inside the Windows VM to improve performance and enable extra features.
- Turbo C++, a legacy C/C++ compiler, is downloaded from the internet inside the virtual machine.
- The downloaded TURBOC3.zip file is extracted to a simple path for easy access.
- TC.EXE is launched from the BIN folder to open the Turbo C++ editor.
- A sample C++ program is written, saved, compiled (Alt+F9), and executed (Ctrl+F9).
- The correct output confirms that the VM can run application software just like a physical computer.

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NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 4

TEACHER'S SIGNATURE:

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Title: Web Service (JAX-WS) for a Calculator Application

Aim: To develop and deploy a SOAP-based web service using JAX-WS that performs basic arithmetic operations such as addition, subtraction, and multiplication.

Objective: To understand the concept of web services in cloud computing, and to learn how to create, publish, and invoke a SOAP web service using Java (JAX-WS).

File Name : CalculatorService04

```
package com.demo.ws04;

import javax.xml.ws.WebService;
import javax.xml.ws.WebMethod;
import javax.xml.ws.WebParam;

@WebService(serviceName = "Calculatorservice04")
public class Calculatorservice04 {

    @WebMethod
    public int add_number(int a, int b)
    {
        return a+b;
    }

    @WebMethod
    public int sub_number(int a, int b)
    {
        return a-b;
    }

    @WebMethod
    public int mult_number(int a, int b)
    {
        return a*b;
    }
}
```


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Explanation:

- A web service exposes functionality over a network so it can be accessed by different applications regardless of platform.
- The class Calculatorservice04 is annotated with @WebService to mark it as a web service.
- Its methods (add_number, sub_number, mult_number) are annotated with @WebMethod to expose them as service operations.
- Each method accepts two integers and performs the respective arithmetic operation.
- The service is deployed and made accessible through its WSDL for client applications.
- This demonstrates how remote functionality can be consumed as a service in a cloud environment.

Calculatorservice04 Web Service Tester

This form will allow you to test your web service implementation ([WSDL File](#))

To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name.

Methods :

public abstract int com.demo.ws04.Calculatorservice04.subNumber(int,int)

subNumber (5) (2)

public abstract int com.demo.ws04.Calculatorservice04.addNumber(int,int)

addNumber (8) (2)

public abstract int com.demo.ws04.Calculatorservice04.multNumber(int,int)

multNumber (5) (8)

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Method invocation trace

localhost:8080/com.Dem.ws04/Calcuatorservice047Tester

addNumber Method invocation

Method parameter(s)

Type	Value
int	8
int	2

Method returned

int : "10"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:add_number xmlns:ns2="http://ws04.demo.com/">
      <arg0>8</arg0>
      <arg1>2</arg1>
    </ns2:add_number>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:add_numberResponse xmlns:ns2="http://ws04.demo.com/">
      <return>10</return>
    </ns2:add_numberResponse>
  </S:Body>
</S:Envelope>
```

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Method invocation trace

localhost:8080/com.Dem.ws04/Calculatorservice04?Tester

subNumber Method invocation

Method parameter(s)

Type	Value
int	5
int	2

Method returned

int : "3"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:sub_number xmlns:ns2="http://ws04.demo.com/">
      <arg0>5</arg0>
      <arg1>2</arg1>
    </ns2:sub_number>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:sub_numberResponse xmlns:ns2="http://ws04.demo.com/">
      <return>3</return>
    </ns2:sub_numberResponse>
  </S:Body>
</S:Envelope>
```

Method invocation trace

localhost:8080/com.Dem.ws04/Calculatorservice04?Tester

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Web Services	
Endpoint	Information
Service Name: {http://ws04.demo.com/}Calculatorservice04 Port Name: {http://ws04.demo.com/}Calculatorservice04Port	Address: http://col-bscit-pc5:8080/com.Dem.ws04/Calculatorservice04 WSDL: http://col-bscit-pc5:8080/com.Dem.ws04/Calculatorservice04?wsdl Implementation class: com.demo.ws04.Calculatorservice04

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FY / SY / TY B. Sc (I. T.) Semester_____

NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 5

TEACHER'S SIGNATURE:

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Title: Currency Converter – RESTful Web Service

Aim: To develop a RESTful web service that converts an amount in Indian Rupees to US Dollars, and to access it through multiple client applications (Java, Web, Python, and Node.js).

Objective: To understand the concept of RESTful web services and how they can be consumed by heterogeneous clients built in different technologies over HTTP.

File Name: CurrencyServer.java

```
import com.sun.net.httpserver.HttpServer;

import com.sun.net.httpserver.HttpHandler;

import com.sun.net.httpserver.HttpExchange;

import java.io.IOException;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.URI;

import java.util.HashMap;

import java.util.Map;

public class CurrencyServer {

    private static final double INR_TO_USD_RATE = 95.00;

    public static void main(String[] args) throws IOException {

        // Create HTTP server on port 8080

        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);

        // Bind the endpoint /api/convert

        server.createContext("/api/convert", new ConvertHandler());

        server.setExecutor(null); // Default executor

        System.out.println("Server started on http://localhost:8080/api/convert");

        server.start();
```

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```
}
```

```
static class ConvertHandler implements HttpHandler {
```

```
    @Override
```

```
    public void handle(HttpExchange exchange) throws IOException {
```

```
        // Set CORS headers so web browsers can call it easily
```

```
        exchange.getResponseHeaders().add("Access-Control-Allow-Origin", "*");
```

```
        exchange.getResponseHeaders().add("Content-Type", "application/json");
```

```
        if ("GET".equalsIgnoreCase(exchange.getRequestMethod())) {
```

```
            URI requestURI = exchange.getRequestURI();
```

```
            String query = requestURI.getQuery();
```

```
            Map<String, String> queryParams = parseQueryParams(query);
```

```
            double rupees = 0.0;
```

```
            if (queryParams.containsKey("rupees")) {
```

```
                try {
```

```
                    rupees = Double.parseDouble(queryParams.get("rupees"));
```

```
                } catch (NumberFormatException e) {
```

```
                    rupees = 0.0;
```

```
                }
```

```
            }
```

```
            double dollars = rupees * INR_TO_USD_RATE;
```

```
            double roundedDollars = Math.round(dollars); // 100.0;
```

```
            // Construct raw JSON response
```

```
            String jsonResponse = String.format(
```

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```
"{"inputRupees\: %.2f, \"convertedDollars\: %.2f, \"exchangeRateUsed\: %.3f,
\"status\: \"SUCCESS\"}";
```

```
rupees, roundedDollars, INR_TO_USD_RATE
```

```
);
```

```
exchange.sendResponseHeaders(200, jsonResponse.length());
```

```
OutputStream os = exchange.getResponseBody();
```

```
os.write(jsonResponse.getBytes());
```

```
os.close();
```

```
} else {
```

```
// Method Not Allowed
```

```
exchange.sendResponseHeaders(405, -1);
```

```
}
```

```
}
```

```
// Utility method to parse query parameters (?rupees=5000)
```

```
private Map<String, String> parseQueryParams(String query) {
```

```
    Map<String, String> result = new HashMap<>();
```

```
    if (query == null) return result;
```

```
    for (String param : query.split("&")) {
```

```
        String[] pair = param.split("=");
```

```
        if (pair.length > 1) {
```

```
            result.put(pair[0], pair[1]);
```

```
        } else if (pair.length == 1) {
```

```
            result.put(pair[0], "");
```

```
        }
```

```
    }
```


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```
        return result;

    }

}

}
```

File Name: ClientApp.java

```
import java.net.URI;

import java.net.http.HttpClient;

import java.net.http.HttpRequest;

import java.net.http.HttpResponse;

import java.util.Scanner;

public class ClientApp {

    public static void main(String[] args) {

        try {

            Scanner scan=new Scanner(System.in);

            System.out.println("Enter USD Amount");

            double inputRupees = scan.nextDouble();

            String url = "http://localhost:8080/api/convert?rupees=" + inputRupees;

            HttpClient client = HttpClient.newHttpClient();

            HttpRequest request = HttpRequest.newBuilder()

                .uri(URI.create(url))

                .GET()

                .build();

            HttpResponse<String> response = client.send(request,

                HttpResponse.BodyHandlers.ofString());

            System.out.println("HTTP Status Code: " + response.statusCode());

        }

    }

}
```

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```
        System.out.println("Response Payload: " + response.body());

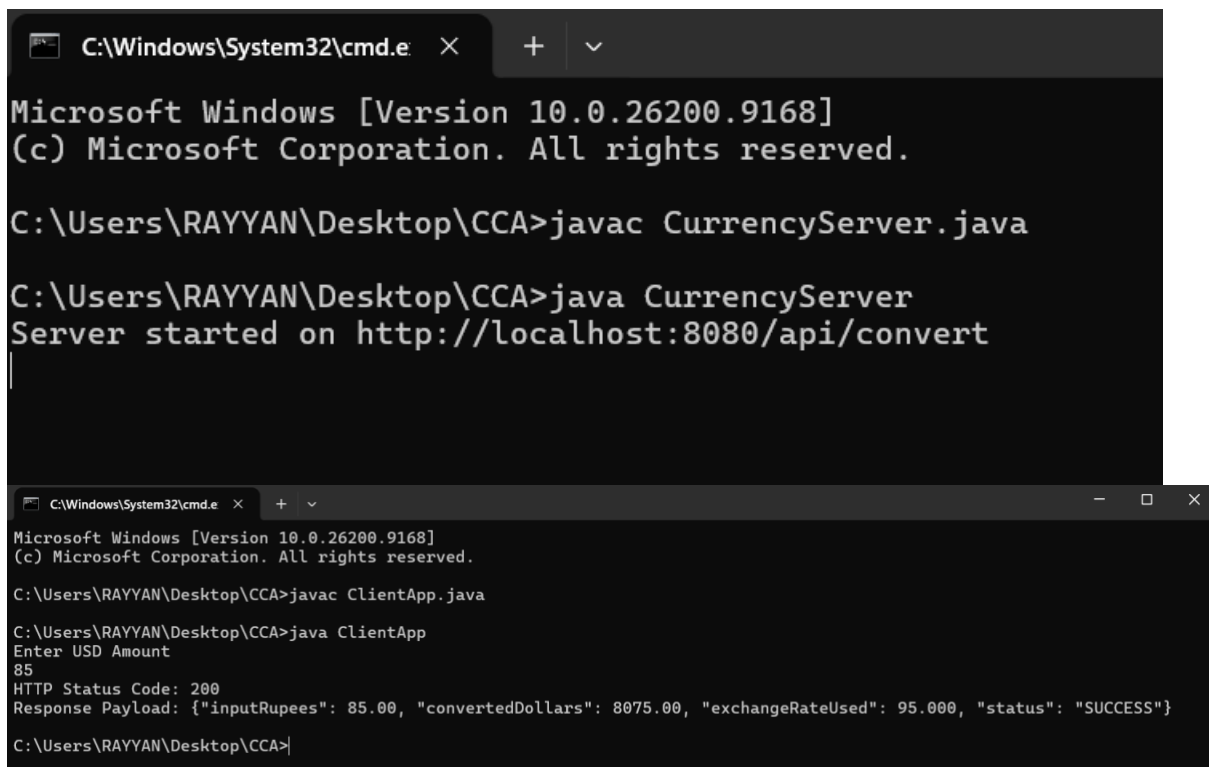
    } catch (Exception e) {

        e.printStackTrace();

    }

}

}
```



The image shows two screenshots of a Windows command prompt window. The top screenshot shows the compilation and execution of a Java program. The bottom screenshot shows the execution of a client application that interacts with a server.

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN\Desktop\CCA>javac CurrencyServer.java

C:\Users\RAYYAN\Desktop\CCA>java CurrencyServer
Server started on http://localhost:8080/api/convert

C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN\Desktop\CCA>javac ClientApp.java

C:\Users\RAYYAN\Desktop\CCA>java ClientApp
Enter USD Amount
85
HTTP Status Code: 200
Response Payload: {"inputRupees": 85.00, "convertedDollars": 8075.00, "exchangeRateUsed": 95.000, "status": "SUCCESS"}

C:\Users\RAYYAN\Desktop\CCA>
```

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5a) Web Based

File Name :index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Currency Converter</title>

  <style>

    body { font-family: Arial, sans-serif; margin: 40px; max-width: 400px; }

    .card { border: 1px solid #ccc; padding: 20px; border-radius: 8px; }

    input, button { width: 100%; padding: 10px; margin-top: 10px; box-sizing: border-box; }

    button { background-color: #007bff; color: white; border: none; cursor: pointer; }

    #result { margin-top: 15px; font-weight: bold; }

  </style>

</head>

<body>

  <div class="card">

    <h2>Currency Converter</h2>

    <label for="rupees">Enter Amount in Rupees (INR):</label>

    <input type="number" id="rupees" placeholder="e.g. 5000">

    <button onclick="convertCurrency()">Convert</button>

    <div id="result"></div>

  </div>
```

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```
<script>

    async function convertCurrency() {

        const rupees = document.getElementById('rupees').value;

        const resultDiv = document.getElementById('result');

        if (!rupees) {

            resultDiv.innerText = "Please enter an amount.";

            return;

        }

        try {

            const response = await fetch(`http://localhost:8080/api/convert?rupees=${rupees}`);

            const data = await response.json();

            resultDiv.innerHTML = `

                Status: ${data.status}<br>

                Rupees: ₹${data.inputRupees}<br>

                Converted USD: $$${data.convertedDollars}<br>

                Rate: ${data.exchangeRateUsed};

            `

        } catch (error) {

            resultDiv.innerText = "Error connecting to server.";

        }

    }

</script>

</body>

</html>
```

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File Name : CurrencyServer.java

```
import com.sun.net.httpserver.HttpServer;

import com.sun.net.httpserver.HttpHandler;

import com.sun.net.httpserver.HttpExchange;

import java.io.IOException;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.URI;

import java.util.HashMap;

import java.util.Map;

public class CurrencyServer {

    private static final double INR_TO_USD_RATE = 95.00;

    public static void main(String[] args) throws IOException {

        // Create HTTP server on port 8080

        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);

        // Bind the endpoint /api/convert

        server.createContext("/api/convert", new ConvertHandler());

        server.setExecutor(null); // Default executor

        System.out.println("Server started on http://localhost:8080/api/convert");

        server.start();

    }

    static class ConvertHandler implements HttpHandler {

        @Override

        public void handle(HttpExchange exchange) throws IOException {
```

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```
// Set CORS headers so web browsers can call it easily

exchange.getResponseHeaders().add("Access-Control-Allow-Origin", "*");

exchange.getResponseHeaders().add("Content-Type", "application/json");

if ("GET".equalsIgnoreCase(exchange.getRequestMethod())) {

    URI requestURI = exchange.getRequestURI();

    String query = requestURI.getQuery();

    Map<String, String> queryParams = parseQueryParams(query);

    double rupees = 0.0;

    if (queryParams.containsKey("rupees")) {

        try {

            rupees = Double.parseDouble(queryParams.get("rupees"));

        } catch (NumberFormatException e) {

            rupees = 0.0;

        }

    }

    double dollars = rupees * INR_TO_USD_RATE;

    double roundedDollars = Math.round(dollars); // 100.0;

    // Construct raw JSON response

    String jsonResponse = String.format(

        "{\"inputRupees\": %.2f, \"convertedDollars\": %.2f, \"exchangeRateUsed\": %.3f, \"status\": \"SUCCESS\"}",

        rupees, roundedDollars, INR_TO_USD_RATE

    );

    exchange.sendResponseHeaders(200, jsonResponse.length());

    OutputStream os = exchange.getResponseBody();
```

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```
os.write(jsonResponse.getBytes());

os.close();

} else {

    // Method Not Allowed

    exchange.sendResponseHeaders(405, -1);

}

}

// Utility method to parse query parameters (?rupees=5000)

private Map<String, String> parseQueryParams(String query) {

    Map<String, String> result = new HashMap<>();

    if (query == null) return result;

    for (String param : query.split("&")) {

        String[] pair = param.split("=");

        if (pair.length > 1) {

            result.put(pair[0], pair[1]);

        } else if (pair.length == 1) {

            result.put(pair[0], "");

        }

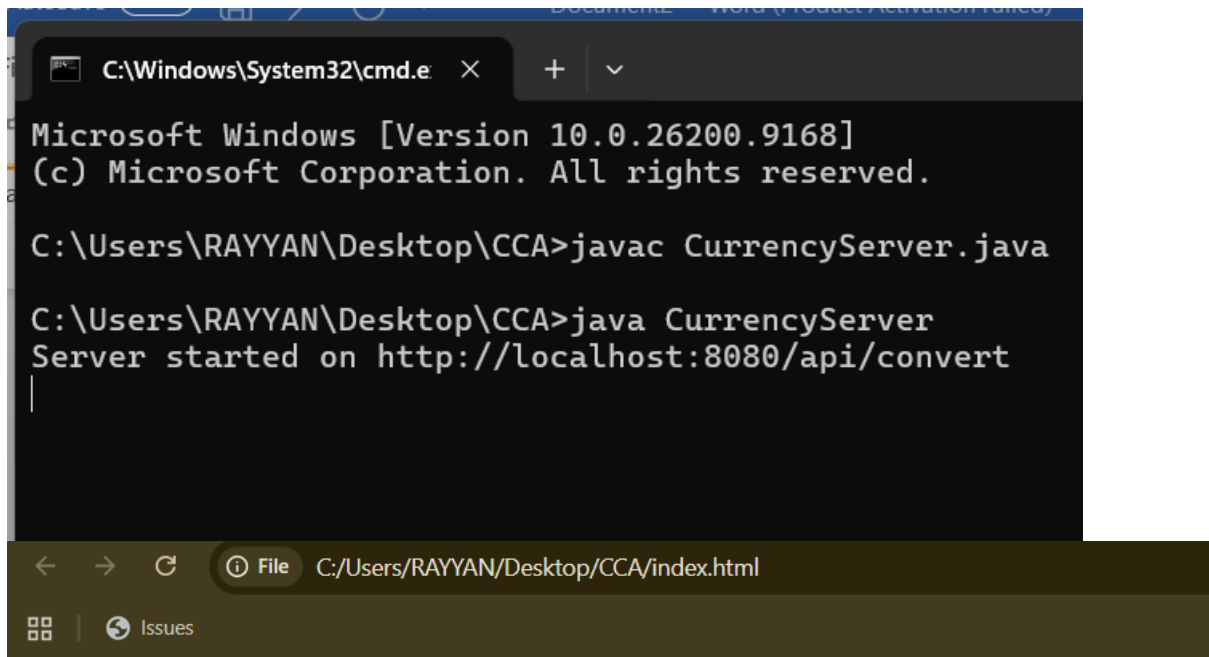
    }

    return result;

}

}
```

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The screenshot shows a Windows command prompt window with the following text:

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN\Desktop\CCA>javac CurrencyServer.java

C:\Users\RAYYAN\Desktop\CCA>java CurrencyServer
Server started on http://localhost:8080/api/convert
```

Below the command prompt, a web browser window is visible, showing the file path `C:/Users/RAYYAN/Desktop/CCA/index.html` and a status bar with the text "Issues".

Currency Converter

Enter Amount in Rupees (INR):

Convert

Status: SUCCESS
Rupees: ₹58
Converted USD: \$5510
Rate: 95

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5B) Python

File Name: CurrencyServer.java

```
import com.sun.net.httpserver.HttpServer;

import com.sun.net.httpserver.HttpHandler;

import com.sun.net.httpserver.HttpExchange;

import java.io.IOException;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.URI;

import java.util.HashMap;

import java.util.Map;

public class CurrencyServer {

    private static final double INR_TO_USD_RATE = 95.00;

    public static void main(String[] args) throws IOException {

        // Create HTTP server on port 8080

        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);

        // Bind the endpoint /api/convert

        server.createContext("/api/convert", new ConvertHandler());

        server.setExecutor(null); // Default executor

        System.out.println("Server started on http://localhost:8080/api/convert");

        server.start();

    }

    static class ConvertHandler implements HttpHandler {
```

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@Override

```
public void handle(HttpExchange exchange) throws IOException {
```

```
    // Set CORS headers so web browsers can call it easily
```

```
    exchange.getResponseHeaders().add("Access-Control-Allow-Origin", "*");
```

```
    exchange.getResponseHeaders().add("Content-Type", "application/json");
```

```
    if ("GET".equalsIgnoreCase(exchange.getRequestMethod())) {
```

```
        URI requestURI = exchange.getRequestURI();
```

```
        String query = requestURI.getQuery();
```

```
        Map<String, String> queryParams = parseQueryParams(query);
```

```
        double rupees = 0.0;
```

```
        if (queryParams.containsKey("rupees")) {
```

```
            try {
```

```
                rupees = Double.parseDouble(queryParams.get("rupees"));
```

```
            } catch (NumberFormatException e) {
```

```
                rupees = 0.0;
```

```
            }
```

```
        }
```

```
        double dollars = rupees * INR_TO_USD_RATE;
```

```
        double roundedDollars = Math.round(dollars); // 100.0;
```

```
        // Construct raw JSON response
```

```
        String jsonResponse = String.format(
```

```
            "{\"inputRupees\": %.2f, \"convertedDollars\": %.2f, \"exchangeRateUsed\": %.3f, \"status\": \"SUCCESS\"}",
```

```
            rupees, roundedDollars, INR_TO_USD_RATE
```

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```
);

exchange.sendResponseHeaders(200, jsonResponse.length());

OutputStream os = exchange.getResponseBody();

    os.write(jsonResponse.getBytes());

    os.close();

} else {

    // Method Not Allowed

    exchange.sendResponseHeaders(405, -1);

}

}

// Utility method to parse query parameters (?rupees=5000)

private Map<String, String> parseQueryParams(String query) {

    Map<String, String> result = new HashMap<>();

    if (query == null) return result;

    for (String param : query.split("&")) {

        String[] pair = param.split("=");

        if (pair.length > 1) {

            result.put(pair[0], pair[1]);

        } else if (pair.length == 1) {

            result.put(pair[0], "");

        }

    }

    return result;

}

} }
```

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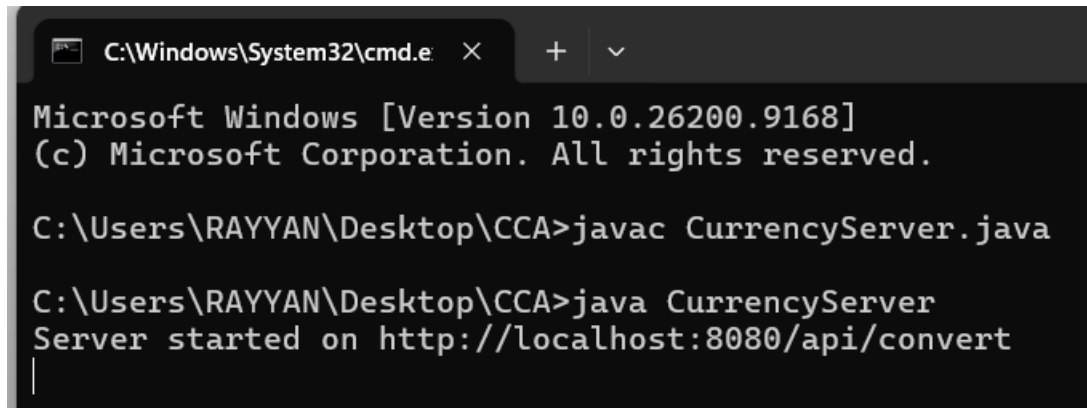
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File Name : client.py

```
import urllib.request
```

```
response = urllib.request.urlopen('http://localhost:8080/api/convert?rupees=5000')
```

```
print(response.read().decode('utf-8'))
```

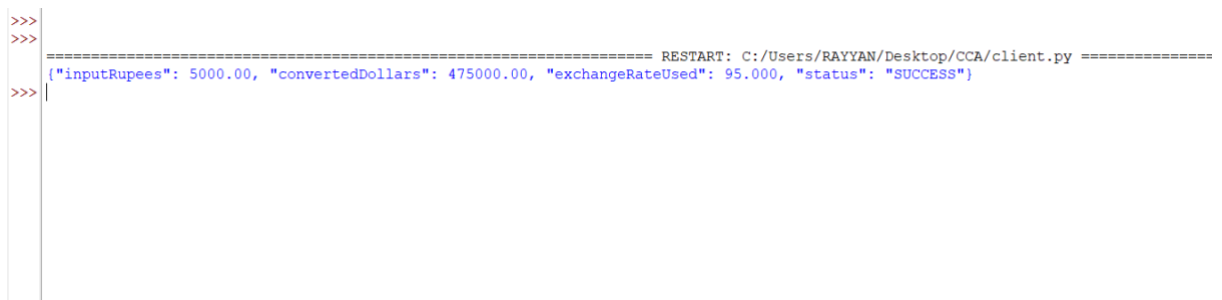


```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN\Desktop\CCA>javac CurrencyServer.java

C:\Users\RAYYAN\Desktop\CCA>java CurrencyServer
Server started on http://localhost:8080/api/convert
|
```



```
>>>
>>> ===== RESTART: C:/Users/RAYYAN/Desktop/CCA/client.py =====
{"inputRupees": 5000.00, "convertedDollars": 475000.00, "exchangeRateUsed": 95.000, "status": "SUCCESS"}
>>> |
```

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5C) Node

```
import com.sun.net.httpserver.HttpServer;

import com.sun.net.httpserver.HttpHandler;

import com.sun.net.httpserver.HttpExchange;

import java.io.IOException;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.URI;

import java.util.HashMap;

import java.util.Map;

public class CurrencyServer {

    private static final double INR_TO_USD_RATE = 95.00;

    public static void main(String[] args) throws IOException {

        // Create HTTP server on port 8080

        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);

        // Bind the endpoint /api/convert

        server.createContext("/api/convert", new ConvertHandler());

        server.setExecutor(null); // Default executor

        System.out.println("Server started on http://localhost:8080/api/convert");

        server.start();

    }

    static class ConvertHandler implements HttpHandler {

        @Override
```

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```
public void handle(HttpExchange exchange) throws IOException {

    // Set CORS headers so web browsers can call it easily

    exchange.getResponseHeaders().add("Access-Control-Allow-Origin", "*");

    exchange.getResponseHeaders().add("Content-Type", "application/json");

    if ("GET".equalsIgnoreCase(exchange.getRequestMethod())) {

        URI requestURI = exchange.getRequestURI();

        String query = requestURI.getQuery();

        Map<String, String> queryParams = parseQueryParams(query);

        double rupees = 0.0;

        if (queryParams.containsKey("rupees")) {

            try {

                rupees = Double.parseDouble(queryParams.get("rupees"));

            } catch (NumberFormatException e) {

                rupees = 0.0;

            }

        }

        double dollars = rupees * INR_TO_USD_RATE;

        double roundedDollars = Math.round(dollars); // 100.0;

        // Construct raw JSON response

        String jsonResponse = String.format(

            "{\"inputRupees\": %.2f, \"convertedDollars\": %.2f, \"exchangeRateUsed\": %.3f, \"status\": \"SUCCESS\"}",

            rupees, roundedDollars, INR_TO_USD_RATE

        );

        exchange.sendResponseHeaders(200, jsonResponse.length());

    }

}
```

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```
OutputStream os = exchange.getResponseBody();

    os.write(jsonResponse.getBytes());

    os.close();

} else {

    // Method Not Allowed

    exchange.sendResponseHeaders(405, -1);

}

}

// Utility method to parse query parameters (?rupees=5000)
private Map<String, String> parseQueryParams(String query) {

    Map<String, String> result = new HashMap<>();

    if (query == null) return result;

    for (String param : query.split("&")) {

        String[] pair = param.split("=");

        if (pair.length > 1) {

            result.put(pair[0], pair[1]);

        } else if (pair.length == 1) {

            result.put(pair[0], "");

        }

    }

    return result;

}

}

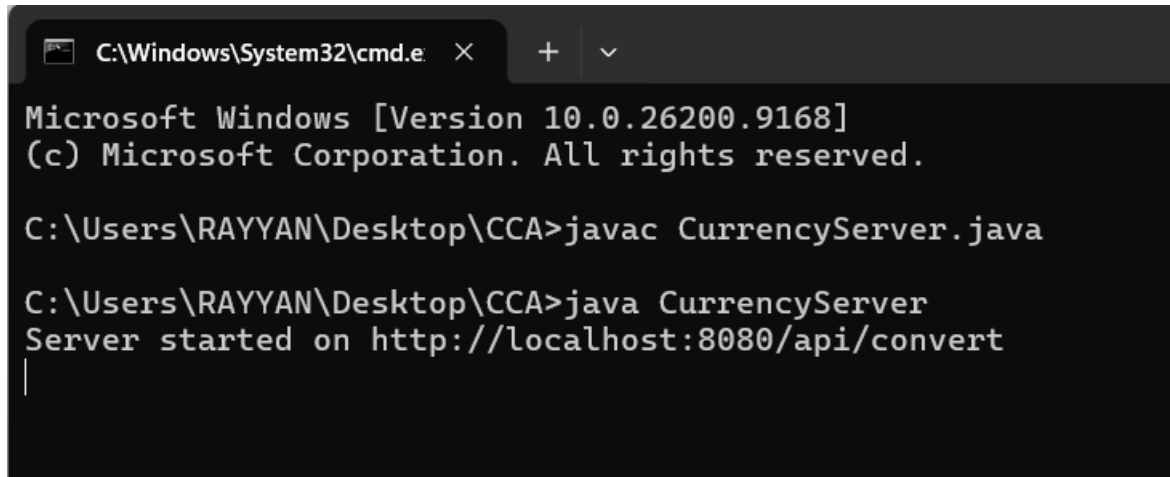
}
```

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Node.js

```
node -e "fetch('http://localhost:8080/api/convert?rupees=5000').then(r => r.json()).then(console.log)"
```

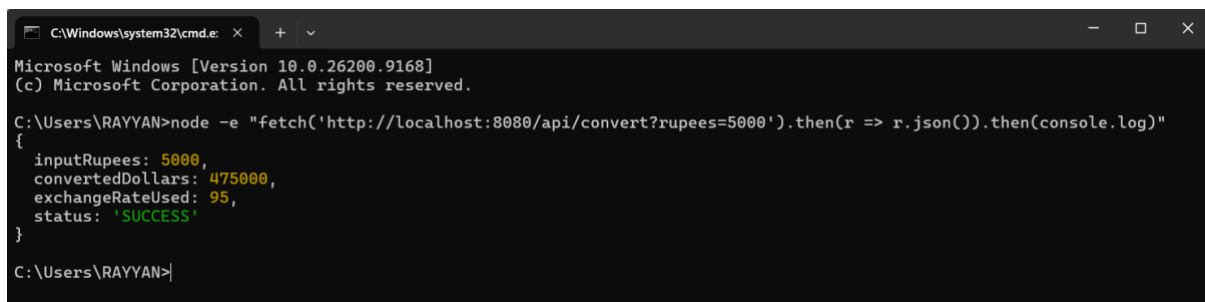


```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN\Desktop\CCA>javac CurrencyServer.java

C:\Users\RAYYAN\Desktop\CCA>java CurrencyServer
Server started on http://localhost:8080/api/convert
|
```



```
C:\Windows\system32\cmd.e:  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAYYAN>node -e "fetch('http://localhost:8080/api/convert?rupees=5000').then(r => r.json()).then(console.log)"
{
  inputRupees: 5000,
  convertedDollars: 475000,
  exchangeRateUsed: 95,
  status: 'SUCCESS'
}

C:\Users\RAYYAN>
```

Explanation:

- REST is an architectural style for building lightweight, stateless web services that communicate over HTTP.
- A CurrencyServer is created using Java's built-in HttpServer class to host the service.
- The server exposes an /api/convert endpoint that accepts the rupee amount as a query parameter.
- The server calculates the converted dollar amount and returns it as a JSON response.
- The service is tested using multiple clients — a Java console client, a Python client, a Node.js client, and a web-based HTML/JavaScript client.
- Each client sends an HTTP GET request and displays the converted result, showing platform-independent access to the service.

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NAME : ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 6

TEACHER'S SIGNATURE:

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FY / SY / TY B. Sc (I. T.) Semester_____

Title: RESTful Web Service with Multiple Endpoints

Aim: To develop a REST-based web service exposing multiple endpoints (/api/hello and /api/status) using Java's HttpServer.

Objective: To understand how multiple resources/routes can be exposed through a single REST service and how each endpoint returns an appropriate JSON response.

File Name : RestServer

```
import com.sun.net.httpserver.HttpServer;
import com.sun.net.httpserver.HttpHandler;
import com.sun.net.httpserver.HttpExchange;
import java.io.IOException;
import java.io.OutputStream;
import java.net.InetSocketAddress;

public class RestServer {

    public static void main(String[] args) throws IOException {

        // Start server on port 8080 with default backlog
        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);

        // Define endpoints (routes)
        server.createContext("/api/hello", new HelloHandler());
        server.createContext("/api/status", new StatusHandler());

        // Use default executor for handling threads
        server.setExecutor(null);

        System.out.println("REST Service started on port 8080...");

        server.start();
    }

    // Handles: GET http://localhost:8080/api/hello
    static class HelloHandler implements HttpHandler {

        @Override
        public void handle(HttpExchange exchange) throws IOException {

            if ("GET".equals(exchange.getRequestMethod())) {

                String response = "{\"message\": \"Hello,Kaushal\"}";
```

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```
        sendResponse(exchange, response, 200, "application/json");
    } else {
        sendResponse(exchange, "Method Not Allowed", 450, "text/plain");
    }
}
}

// Handles: GET http://localhost:8080/api/status
static class StatusHandler implements Handler {
    @Override
    public void handle(HttpExchange exchange) throws IOException {
        String response = "{\"status\": \"UP\", \"database\": \"Connected\"}";
        sendResponse(exchange, response, 200, "application/json");
    }
}

// Helper method to send JSON or Text HTTP responses
private static void sendResponse(HttpExchange exchange, String response, int statusCode,
String contentType) throws IOException {
    exchange.getResponseHeaders().set("Content-Type", contentType);
    exchange.sendResponseHeaders(statusCode, response.getBytes().length);
    OutputStream os = exchange.getResponseBody();
    os.write(response.getBytes());
    os.close();
}
}
```

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```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

D:\CCA>javac RestServer.java

D:\CCA>java RestServer
REST Service started on port 8080...
|
```

```
WhatsApp X localhost:8080/api/hello X +
localhost:8080/api/hello
pretty-print ☐
{"message": "Hello,Kaushal"}
```

```
WhatsApp X localhost:8080/api/status X +
localhost:8080/api/status
Pretty-print ☐
{"status": "UP", "database": "Connected"}
```

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Explanation:

- In REST architecture, different resources are represented by different URIs (endpoints).
- A RestServer is implemented using Java's HttpServer class to host the service.
- Two contexts are registered: /api/hello and /api/status.
- The /api/hello endpoint is handled by HelloHandler, which returns a greeting message in JSON.
- The /api/status endpoint is handled by StatusHandler, which returns the service's current status.
- A common sendResponse() helper method is used to send JSON responses back to the client for both endpoints.

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FY / SY / TY B. Sc (I. T.) Semester_____

NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 7

TEACHER'S SIGNATURE:

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Title: Java RMI (Remote Method Invocation) Application

Aim: To create a Java RMI application in which a client invokes a method on a remote server to add two numbers.

Objective: To understand the concept of Remote Method Invocation (RMI) and how it enables distributed computing by allowing an object on one machine to invoke methods on an object running on another machine

Calculator.java:

```
import java.rmi.Remote;

import java.rmi.RemoteException;

public interface Calculator extends Remote {

    int add(int firstNumber, int secondNumber) throws RemoteException;

}
```

File Name: CalculatorImpl.java

```
import java.rmi.RemoteException;

import java.rmi.server.UnicastRemoteObject;

public class CalculatorImpl extends UnicastRemoteObject

    implements Calculator {

    public CalculatorImpl() throws RemoteException {

        super();

    }

    @Override

    public int add(int firstNumber, int secondNumber)

        throws RemoteException {

        return firstNumber + secondNumber;

    }

}
```

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File Name: RMI Server.java

```
import java.rmi.registry.LocateRegistry;

import java.rmi.registry.Registry;

public class RMIServer {

    public static void main(String[] args) {

        try {

            Calculator calculator = new CalculatorImpl();

            // Create the RMI registry on port 1099.

            Registry registry = LocateRegistry.createRegistry(1099);

            // Register the remote object.

            registry.rebind("CalculatorService", calculator);

            System.out.println("RMI server is running...");

        } catch (Exception exception) {

            System.err.println("Server error: "

                + exception.getMessage());

            exception.printStackTrace();

        }

    }

}
```


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File Name: RMIClient.java:

```
import java.rmi.registry.LocateRegistry;

import java.rmi.registry.Registry;

import java.util.Scanner;

public class RMIClient {

    public static void main(String[] args) {

        try (Scanner scanner = new Scanner(System.in)) {

            String serverAddress = "localhost";

            Registry registry = LocateRegistry.getRegistry(serverAddress, 1099);

            Calculator calculator = (Calculator)

                registry.lookup("CalculatorService");

            System.out.print("Enter the first number: ");

            int firstNumber = scanner.nextInt();

            System.out.print("Enter the second number: ");

            int secondNumber = scanner.nextInt();

            int result = calculator.add(firstNumber, secondNumber);

            System.out.println("Result received from server: " + result);

        } catch (Exception exception) {

            System.err.println("Client error: "

                + exception.getMessage());

            exception.printStackTrace();

        }

    }

}
```

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```
C:\Windows\System32\cmd.exe - java RMIServer
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>javac RMIServer.java

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>java RMIServer
RMI server is running...
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>javac RMIClient.java

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>java RMIClient
Enter the first number: 2
Enter the second number: 5
Result received from server: 7

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>
```

Explanation:

- RMI allows a Java object in one JVM to invoke methods on an object running in another JVM.
- A remote interface, Calculator, is defined declaring the add() method.
- CalculatorImpl implements the Calculator interface and provides the actual logic for addition.
- RMIServer creates an RMI registry on port 1099 and binds the CalculatorImpl object under the name "CalculatorService".
- RMIClient looks up the "CalculatorService" in the registry and remotely invokes the add() method with two numbers entered by the user.
- The result computed on the server is returned to the client and displayed, demonstrating remote procedure execution.

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NAME: ANSARI MUARIF

ROLL NO : 3

CLASS: TYB.Sc.IT.

**SUBJECT: Cloud Computing and
Architecture**

PRACTICAL NO: 8

TEACHER'S SIGNATURE:

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Title: TCP Client-Server Socket Programming

Aim: To implement a TCP client-server application in Java where the client sends a message to the server and receives a response.

Objective: To understand socket programming in Java and to learn how connection-oriented communication is established between a client and a server using TCP/IP.

File Name: TCPServer.java

```
import java.io.*;
import java.net.*;

public class TCPServer {

    public static void main(String[] args) {

        int port = 5000;

        System.out.println("Server is starting and listening on port " + port + "...");

        try (ServerSocket serverSocket = new ServerSocket(port)) {

            // Wait and accept incoming client connection

            Socket socket = serverSocket.accept();

            System.out.println("Client connected: " + socket.getRemoteSocketAddress());

            // Setup input stream to read data from client

            InputStream input = socket.getInputStream();

            BufferedReader reader = new BufferedReader(new InputStreamReader(input));

            // Setup output stream to send data to client

            OutputStream output = socket.getOutputStream();

            PrintWriter writer = new PrintWriter(output, true);

            // Read client message

            String clientMessage = reader.readLine();

            System.out.println("Received from client: " + clientMessage);

            // Send response back

            String response = "Hello Client! Your message was received.";

            writer.println(response);
```

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```
System.out.println("Response sent to client.");

// Close connection

socket.close();

System.out.println("Connection closed.");

} catch (IOException e) {

    System.out.println("Server exception: " + e.getMessage());

    e.printStackTrace();

}

}

}
```

File Name : TCPClient.java

```
import java.io.*;

import java.net.*;

public class TCPClient {

    public static void main(String[] args) {

        String hostname = "localhost";

        int port = 5000;

        System.out.println("Connecting to server at " + hostname + ":" + port + "...");

        try (Socket socket = new Socket(hostname, port)) {

            // Setup output stream to send data to server

            OutputStream output = socket.getOutputStream();

            PrintWriter writer = new PrintWriter(output, true);

            // Send a message

            String message = "Hello Server! This is a TCP message.";

            writer.println(message);

            System.out.println("Sent to server: " + message);

            // Setup input stream to read server response

            InputStream input = socket.getInputStream();

            BufferedReader reader = new BufferedReader(new InputStreamReader(input));
```

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```
// Read response

String serverResponse = reader.readLine();

System.out.println("Received from server: " + serverResponse);
} catch (UnknownHostException e) {

    System.out.println("Server not found: " + e.getMessage());
} catch (IOException e) {

    System.out.println("I/O error: " + e.getMessage());
}
}
}
```

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>javac TCPServer.java

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>java TCPServer
Server is starting and listening on port 5000...
Client connected: /127.0.0.1:50352
Received from client: Hello Server! This is a TCP message.
Response sent to client.
Connection closed.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>
```

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>javac TCPClient.java

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>java TCPClient
Connecting to server at localhost:5000...
Sent to server: Hello Server! This is a TCP message.
Received from server: Hello Client! Your message was received.

C:\Users\COL-BSCIT-PC4\Desktop\CCA04>_
```

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Explanation:

- TCP provides reliable, connection-oriented communication between two systems.
- TCPServer creates a ServerSocket on a specified port and waits to accept an incoming client connection.
- TCPClient creates a Socket to connect to the server at the given hostname and port.
- The client sends a message to the server using an output stream (PrintWriter).
- The server reads the client's message and sends back a response using its own output stream.
- The client reads and displays the server's response, completing the two-way communication, after which the connection is closed.